Hardware User Manual

JEPMC Series

JEPMC-AN2900
JEPMC-AN2910
JEPMC-PL2910
JEPMC-IO2310
Σ – V Series

AJINEXTEK AXT

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CONTENS

1. System Overview 5
1.1 MECHATROLINK System Overview 5
1.1.1 MECHATROLINK System 5
1.1.2 Characteristics of MECHATROLINK System 5
1.1.3 MECHATROLINK Transmission Specification 5
1.1.4 Cautions in System Environment Setting 5
1.2 MECHATROLINK System Configuration 6
1.2.1 MECHATROLINK II Compatible Products 6
1.3 Communication Setting for Slave 6
1.3.1 JEPMC-AN2900 / JEPMC-AN2910 / JEPMC-PL2910 Communication Setting 6
1.3.2 JEPMC-IO2310 Communication Setting 7
1.3.3 SGDV-xxx Communication Setting 8
2. Distributed I/O Modules 9
2.1 Common Specification 9
2.1.1 Common Specification 9
2.1.2 Installation 9
2.2 Analog Input Module (±10 V, 4 CH)(AN2900) 12
2.2.1 Dimension and Configuration 12
2.2.2 Performance Specification 15
2.2.3 Circuit Configuration 16
2.2.4 Connection Examples 17
2.3 Analog Output Module (±10 V, 2 CH) (AN2910) 18
2.3.1 Dimension and Configuration 18
2.3.2 Performance Specification 21
2.3.3 Circuit Configuration 22
2.3.4 Circuit Wiring Examples 22
2.4 Pulse Output Module 23
2.4.1 Dimension and Configuration 23
2.4.2 System Setting 27
2.4.3 Specification 28
2.4.4 Examples of External I/O Signal and Connection 34
2.5 SERVOPACK (SGDV-□□□□□□□) 36

2.5.1 SERVOPACK Components 37
2.5.2 SERVOPACK Rating and Specification 37
2.5.3 Examples of Servo System Configuration 43
2.5.4 SERVOPACK Model Names 45
2.5.5 Panel Display 46
2.5.6 Wiring and Connection 48
2.5.7 MECHATROLINK-Ⅱ Settings 59
3. Simple I/O Module 61
3.1 64 I/O Module(IO2310) 61
3.1.1 Dimension and Configuration 61
3.1.2 Performance Specification 64
3.1.3 System Connection 66
4. Connections 75
4.1 MECHATROLINK Connectors 75
4.2 MECHATROLINK Cables 77
5. Appendix A Dimension 79

1. System Overview

1.1 MECHATROLINK System Overview

This chapter examines the MECHATROLINK System and its characteristics.

1.1.1 MECHATROLINK System

MECHATROLINK system basically consists of one MECHATROLINK Master and one or more MECHATROLINK modules.

1.1.2 Characteristics of MECHATROLINK System

-. MECHATROLINK System provides a consistent refresh system that is comparable to the local high speed I/O.

-. MECHATROLINK Module is connectable with a single twisted-pair cable

-. Master can detect any error with the Slave module.

-. Slave module automatically connects to the Network.

1.1.3 MECHATROLINK Transmission Specification

Following Table shows the Transmission Specification of MECHATROLINK II System.

Item	MECHATROLINK II Specification
Transmission Method	MECHATROLINK II
Transmission Connection Form	Bus Type
Transmission Connection	Electric Bus
Transmission Connection	50m
Min. Distance between Modules	0.5m
Transmission Speed	10Mbps
Communication Speed	1msec
Max. Number of Slaves	30 Stations
Transmission Control Method	Cyclic
Error Control	CRC check

1.1.4 Cautions in System Environment Setting

-. Setting must be set to 17 Bytes Mode.

-. Transmission speed must be set to 10Mbps.

-. Standard Cable must be used.

-. Terminating Resistor must be installed.

-. Operation must be made without overlap or duplication of Slave ID.

1.2 MECHATROLINK System Configuration

This chapter examines the Compatible products of MECHATROLINK Ⅱ.

1.2.1 MECHATROLINK Ⅱ Compatible Products

(1) Master Device

Device Name	Model	Transmission	Maker
PCI Board	R1604_MLII	M-Ⅱ	AjinExtek

(2) Slave Devices

Device Name	Model	Transmission	Maker
A/D Module Analog input -10 ~ 10V, 2 channels	JEPMC-AN2900	M-Ⅱ	YASKWA
D/A Module Analog output -10 ~ 10V, 2 channels	JEPMC-AN2910	M-Ⅱ	YASKWA
64-point Module 24VDC, 64-point input, 64-point output(sourcing)	JEPMC-IO2310	M-Ⅱ	YASKWA
Pulse Output Module Pulse output, 2 channels	JEPMC-PL2910	M-Ⅱ	YASKWA
Servo Driver	SGDV-xxx	M-Ⅱ	YASKWA

1.3 Communication Setting for Slave

Basic communication setting must be established as MECHATROLINK Slaves are controlled through communication.

Basic settings for communication are listed below.

-. Communication Speed Setting must be set at 10Mbps.

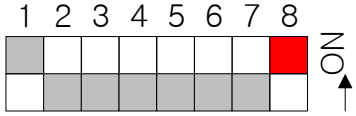
-. Communication Byte must be set at 17Bytes Mode.

-. IDs must not overlap or duplicate.

1.3.1 JEPMC-AN2900 / JEPMC-AN2910 / JEPMC-PL2910 Communication Setting

(1) Communication Speed Setting

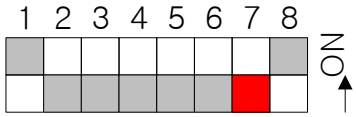
Communication speed must be set at 10Mbps and must set the Dip Switch 8 at “ON” as shown below.



(2) Communication Byte Setting

MECHATROLINK II must be set to either 32 Bytes or 17VBytes Mode. In order to use the Scan in 1mSec with 30 Slaves connected, the Communication Byte Setting must be set at 17Byte Mode.

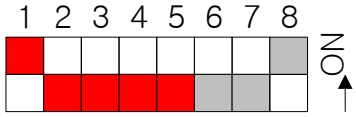
Turn Off the Dip Switch 7 in order to set to 17Bytes Mode



(3) ID Setting

All MECHTROLINK II Slaves must be set with distinct ID and ID Setting is made with Dip Switch.

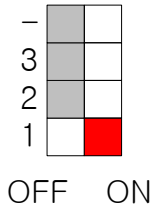
ID Setting is made in a range of 1 to 30 and cannot be set at outside the given range.



1.3.2 JEPMC-IO2310 Communication Setting

(1) Communication Speed Setting

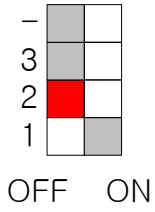
Communication Speed must be set at 10Mbps by setting the Dip Switch 1 at ON as shown below.



(2) Communication Byte Setting

MECHATROLINK II must be set to either 32 Bytes or 17VBytes Mode. In order to use the Scan in 1mSec with 30 Slaves connected, the Communication Byte Setting must be set at 17Byte Mode.

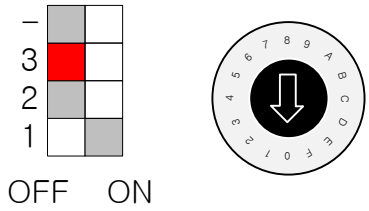
In order to set at 17Bytes Mode, Turn OFF the Dip Switch 2.



(3) ID Setting

All MECHTROLINK II Slaves must have distinct ID, and ID setting must be made using the following switch. In order to set the JEPMC-IO2310 ID, operation of 2 switches are necessary.

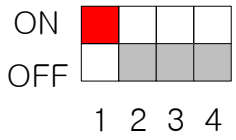
2 Switches are used for ID setting in a range of 1 to 29. If Dip Switch 3 is set at OFF, a range from 1 to 15 can be set. If Dip Switch 3 is set at ON, a range from 16 to 29 can be set.
If Dip Switch 3 is set at OFF, the Rotary Switch must be set in a range from 1 to F. If Dip Switch 3 is set at ON, Rotary Switch must be set in a range from 0 to D.



1.3.3 SGDV-xxx Communication Setting

(1) Communication Setting

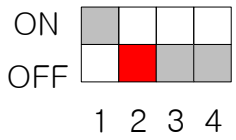
Communication speed must be set at 10 Mbps and Dip Switch 1 must be set at ON.



(2) Communication Byte Setting

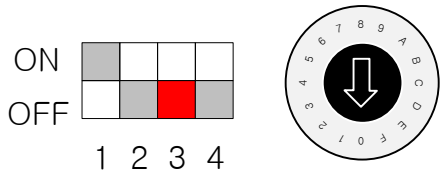
MECHATROLINK II must be set to either 32 Bytes or 17VBytes Mode. In order to use the Scan in 1mSec with 30 Slaves connected, the Communication Byte Setting must be set at 17Byte Mode.

In order to set the Communication Byte Setting at 17Bytes Mode, Set the Dip Switch 2 at OFF.



(3) ID Setting

All MECHTROLINK II Slaves must have distinct ID, and ID setting must be made using the following switch. In order to set the SGDV-xxx, operation of 2 switches are necessary.
2 Switches are used for ID setting in a range of 1 to 29. If Dip Switch 3 is set at OFF, a range from 1 to 15 can be set. If Dip Switch 3 is set at ON, a range from 16 to 29 can be set.
If Dip Switch 3 is set at OFF, the Rotary Switch must be set in a range from 1 to F. If Dip Switch 3 is set at ON, Rotary Switch must be set in a range from 0 to E.



2. Distributed I/O Modules

2.1 Common Specification

This chapter examines Distributed I/O Installation Methods and Common Specification.

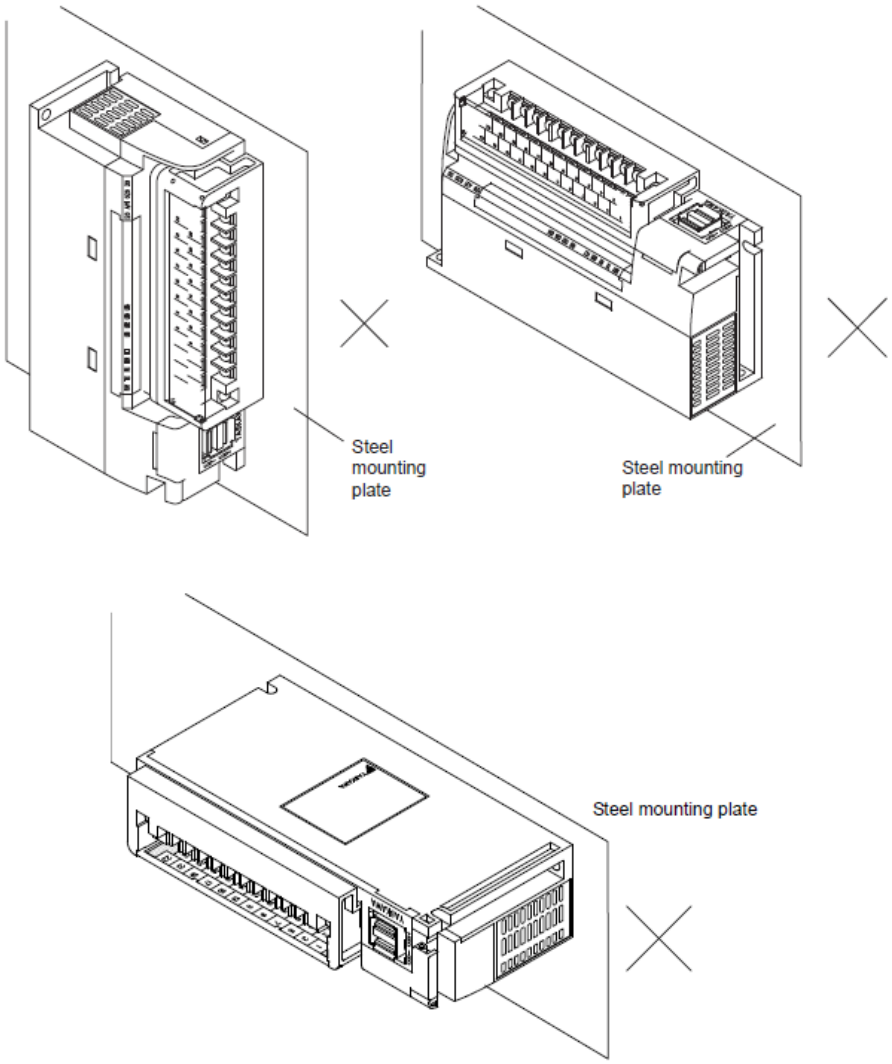
2.1.1 Common Specification

The following table lists the Distributed I/O Specification.

Item		Specification
Main External Power	Rated Voltage	24 VDC
	Range of Allowable Voltage	20.4 ~ 26.4 VDC
	Allowable Ripple	Must be within the range of -15% ~ 10%
Dielectricity		DC/DC Insulation
Dielectric Resistance		Min. 50 MΩ
Environment Factors	Ambient Operating Temperature Range	0 ~ 60 ℃
	Storage Temperature	-25 ~ 85 ℃
	Operating Humidity	30 ~ 90 %
	Storage Humidity	5 ~ 95 %
Conditions for Equipment Operation	Vibration Resistance	10 ~ 57 Hz with half-amplitude of 0.075mm. 57 ~ 150 Hz at fixed acceleration of 9.8 m/s^2. 10 sweep for 8 minutes each in X,Y,Z directions
	Impact Resistance	Peak acceleration of 147 m/s^2 twice for 11 ms in X,Y,Z directions
Conditions for Electric Operation	Noise Resistance	Impulse : ± 1,000 V Fast transient burst : Level 3(1,000 V)

2.1.2 Installation

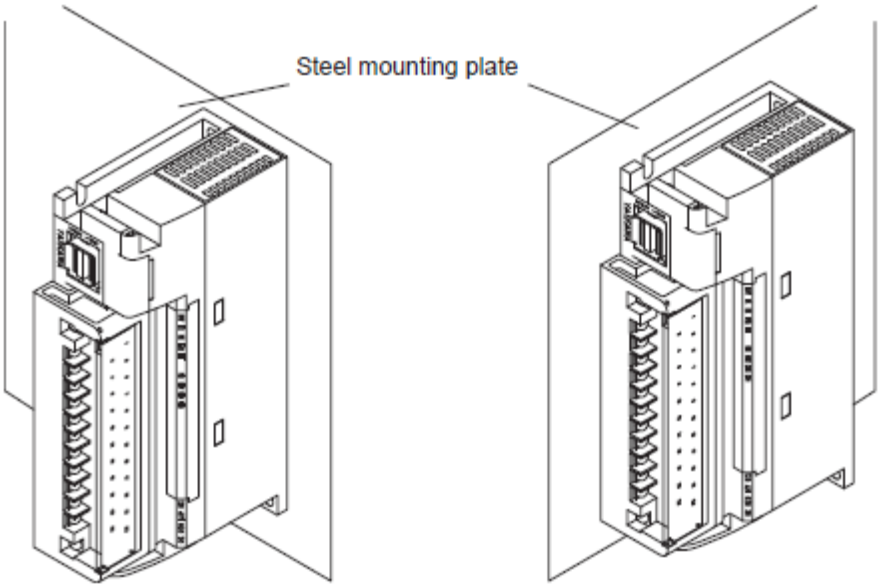
Following images show incorrect methods of installation.



Distributed I/O Module has three methods of installation.

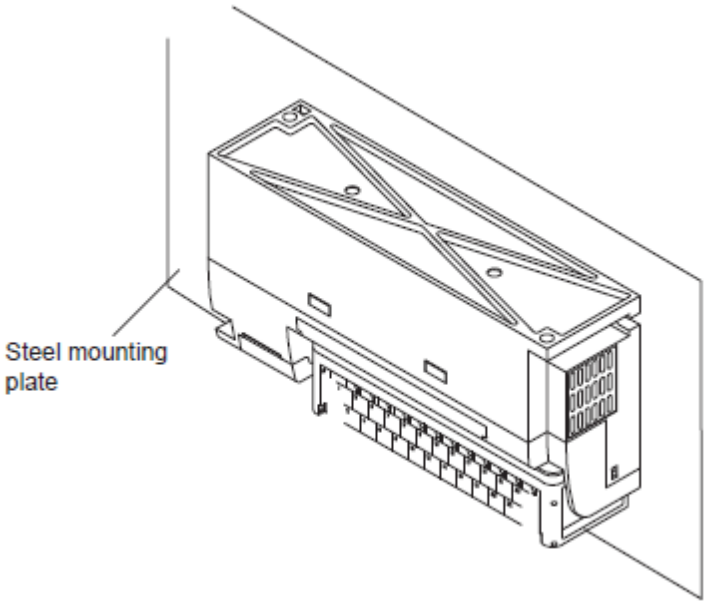
(1) Installation Method 1

If the ambient operating temperature is in range of 0-60℃, the following installation method must be used for installation.



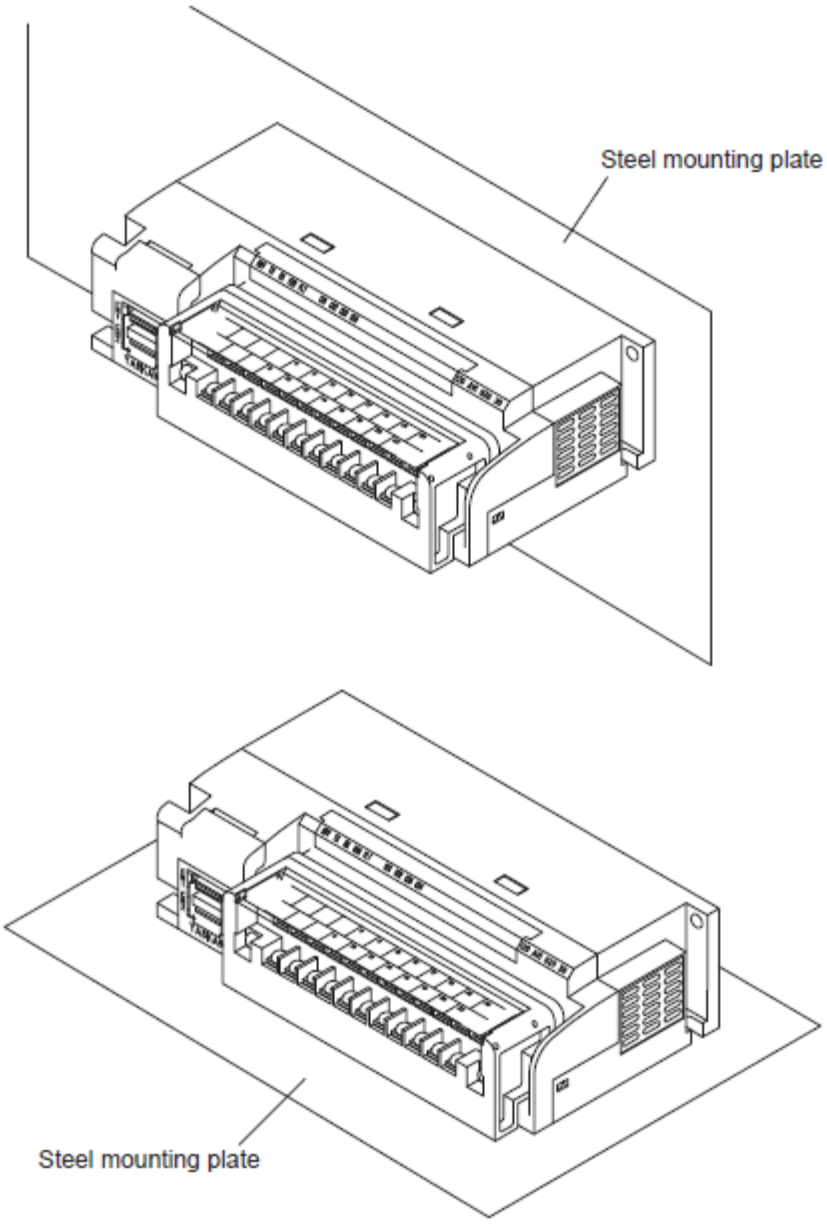
(2) Installation Method 2

If the ambient operating temperature is in range of 0-55℃, the following installation method must be used for installation .



(3) Installation Method 3

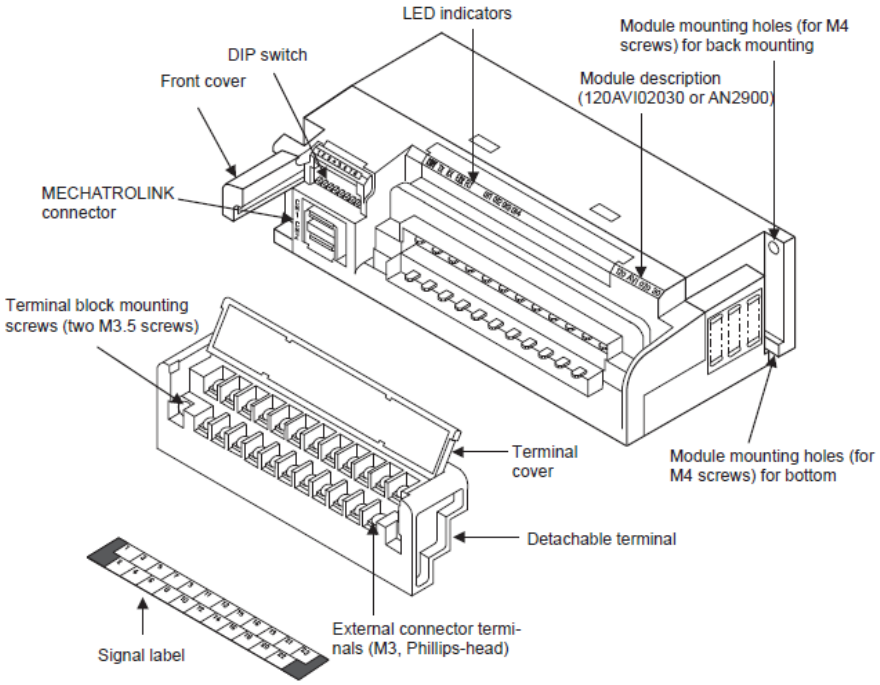
If the ambient operating temperature is in range of 0-55℃, the following installation method must be used for installation.



2.2 Analog Input Module (±10 V, 4 CH)(AN2900)

2.2.1 Dimension and Configuration

The following image shows the dimension of Analog Module.



(1) LED Indication

RDY	TX	RX	ERR	FLT	CH1	CH2	CH3	CH4

Indication Name	Indication Color	Meaning of Lit and Flashing	
RDY	Green	Lit	Normal Status
		Flashing	Cable is cut or awaiting communication from the Master.
TX	Green	Lit	Transmitting Data.
RX	Green	Lit	Receiving Data.
ERR	Red	Lit	Offset/Gain Setting Error
		Flashing	Self-Diagnostic Error
CH1 ~ CH4	Green	Lit	Out of Range (+10.02 V < Input signal -10.02 V > Input signal)

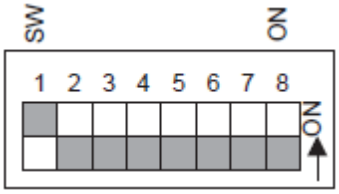
(2) DIP Switch

Before operating the Analog Input Module, the DIP Switch of Module must be set.

(a) DIP Switch Function

DIP Switch consists of a total of 8 pins.

The pins are numbered from 1 to 8 as shown below. Each pin can move up for turn ON.



The following table shows each switch function.

Pin No.	Setting	Function	Factory Setting
1 ~ 5	ON	Set Slave Address	1:ON 2 ~ 5:OFF
	OFF		
6	ON	If SW 8 is set at ON, the communication data length is 32 Bytes. If SW 8 is set at OFF, set SW 6 at OFF.	OFF
	OFF	If SW 8 is set at ON, the communication data length is 17 Bytes. If SW 8 is set at OFF, set SW 6 at OFF.	
7	ON	Software Filter Enable	OFF
	OFF	Software Filter Disable	
8	ON	Communication Speed 10Mbps	ON
	OFF	Communication Speed 4Mbps	

(b) Slave Address Setting

Slave Address of Distributed I/O Module must be set using the DIP Switch in a range of 1-5.

Refer to the following table for setting the slave address.

Pin No					Slave Address
1	2	3	4	5	
0	0	0	0	0	Not used
1	0	0	0	0	1
0	1	0	0	0	2
1	1	0	0	0	3
0	0	1	0	0	4
1	0	1	0	0	5

0	1	1	0	0	6
1	1	1	0	0	7
0	0	0	1	0	8
1	0	0	1	0	9
0	1	0	1	0	10
1	1	0	1	0	11
0	0	1	1	0	12
1	0	1	1	0	13
0	1	1	1	0	14
1	1	1	1	0	15
0	0	0	0	1	16
1	0	0	0	1	17
0	1	0	0	1	18
1	1	0	0	1	19
0	0	1	0	1	20
1	0	1	0	1	21
0	1	1	0	1	22
1	1	1	0	1	23
0	0	0	1	1	24
1	0	0	1	1	25
0	1	0	1	1	26
1	1	0	1	1	27
0	0	1	1	1	28
1	0	1	1	1	29
0	1	1	1	1	30
1	1	1	1	1	Not used

2.2.2 Performance Specification

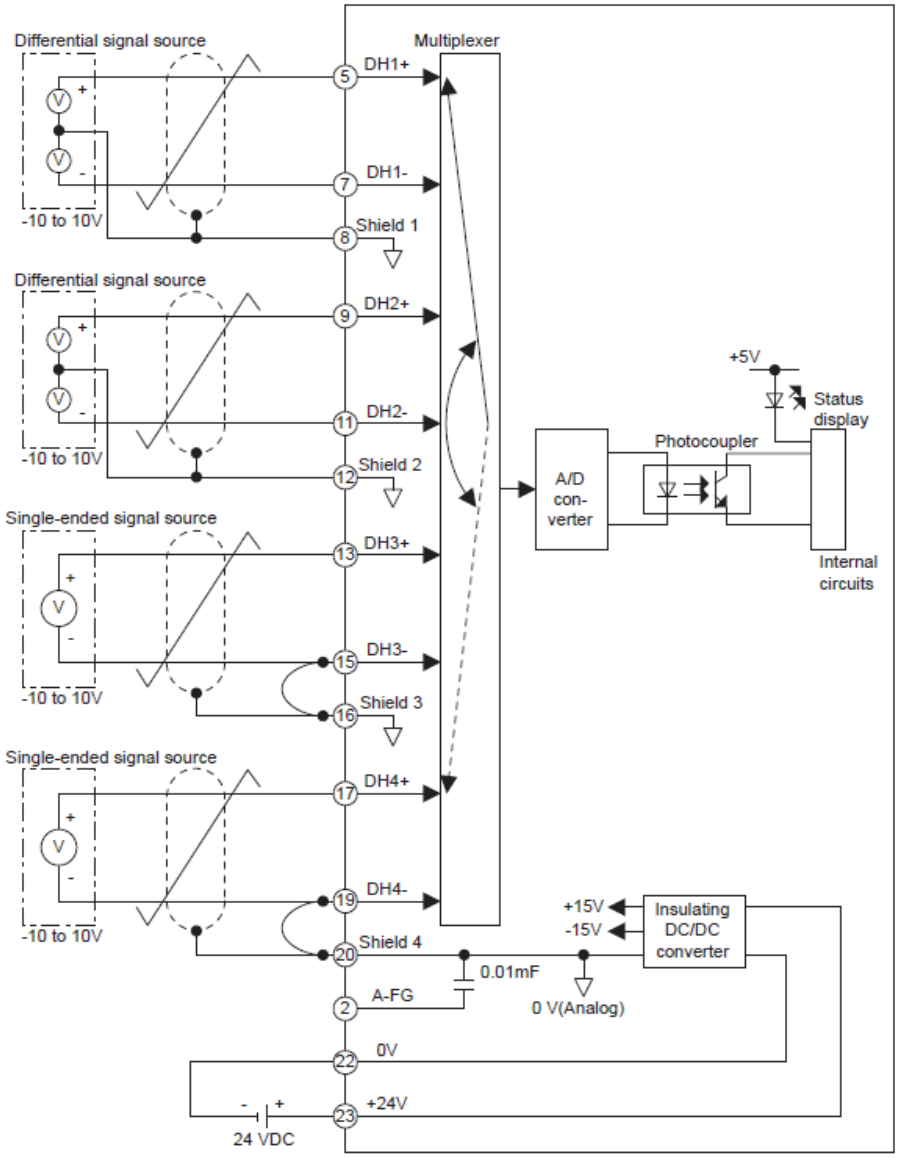
The following table shows the Performance Specification of Analog Input Module (±10 V, 4 CH).

Item	Specification
Name	Analog Input Module (±10 V, 4 CH)
Model Name	AN2900
Model Number	JEPMC-AN2900
Input Signal Range	-10 ~ 10 V
Number of Channels	4

Input Resistance		1 MΩ
Maximum Allowable Input Voltage		-20 ~ 20 V
Digital Resolution		16 bits
Data Form		Binary (2s complement)
Error		±0.5% F.S. (at 25 ℃) ±0.1% F.S. (at 0 ~ 60 ℃)
Input Delay Time		Max. 4ms
Sampling Interval		Communication Cycle
Input Filter		Software Filter
External Connection		Removable Terminal
Status Display		
Input Circuit Dielectricity	Dielectric Method	Photo-Coupler
	Electric Resistance	DC/DC Insulation
	Dielectric Resistance	100 MΩ
External Input Voltage		24 VDC, 120mA
Dimension		161 x 44 x 79 (W x H x D)

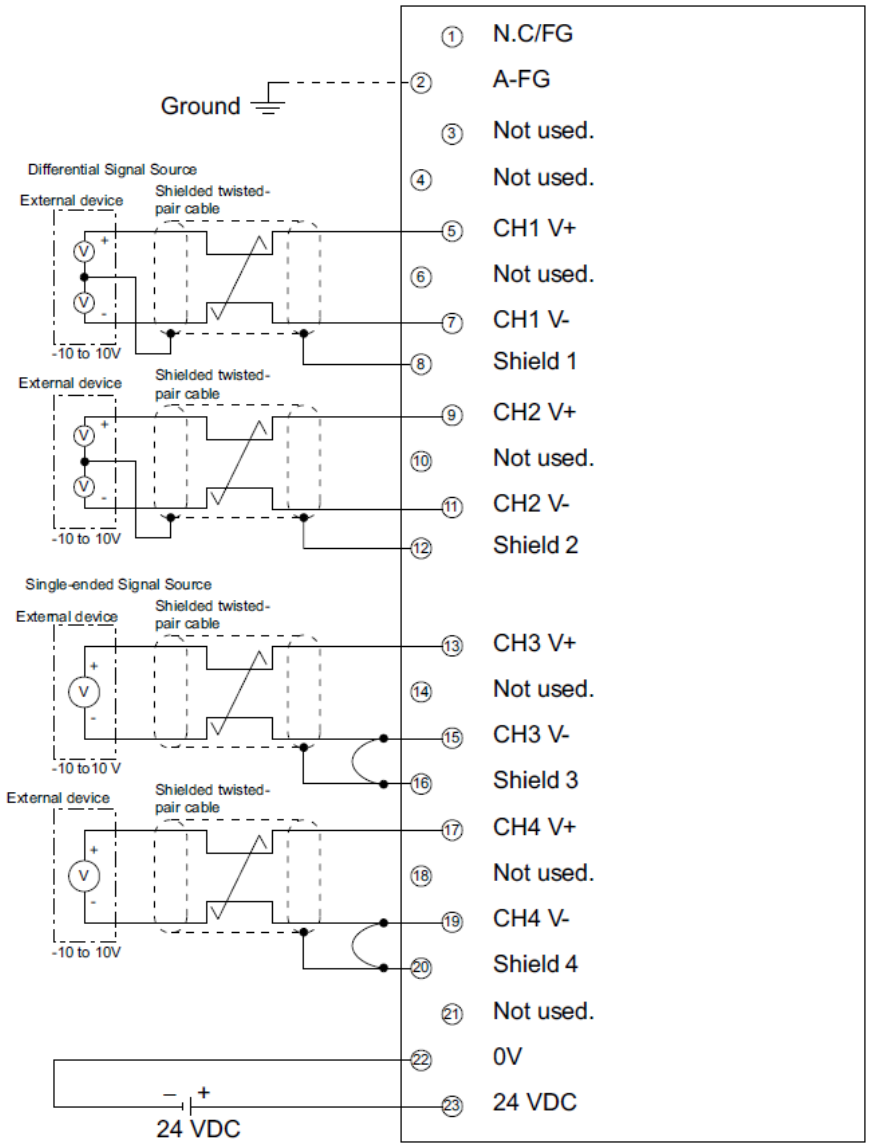
2.2.3 Circuit Configuration

Configure the circuits as shown in the following diagram.



2.2.4 Connection Examples

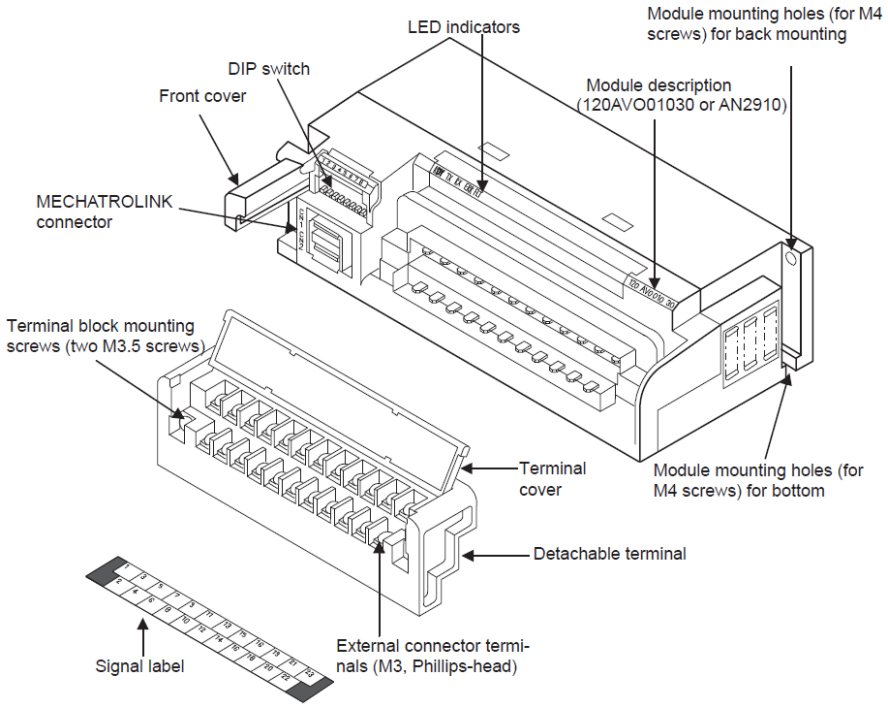
Next diagram shows the examples of connections made terminal block of Analog Input Module.



2.3 Analog Output Module (±10 V, 2 CH) (AN2910)

2.3.1 Dimension and Configuration

Next diagram shows the dimension of Analog Output Module.



(1) LED Indication

RDY TX RX ERR FLT

Indication	Color	Meaning of Lit and Flashing	
RDY	Green	Lit	Norma Status
		Flashing	Cable is cut or awaiting communication from Master.
TX	Green	Lit	Transmitting data.
RX	Green	Lit	Receiving data.
ERR	Red	Lit	Offset/Gain Setting Error
		Flashing	Error Self-Diagnostic Error
CH1 ~ CH4	Green	Lit	Out of Range (+10.02 V < Input signal -10.02 V > Input signal)

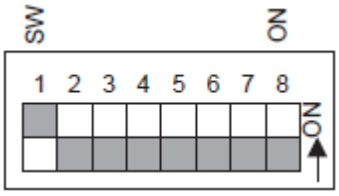
(2) DIP Switch

DIP Switch must be set before operating the Analog Output Module.

(a) DIP Switch Function

DIP Switch consists of a total of 8 pins.

The pins are numbered from 1 to 8 as shown below. Each pin can move up for turn ON.



Next table shows the functions of each switch.

Pin No	Setting	Function	Factory Setting
1 ~ 5	ON	Set Slave Address	1:ON 2 ~ 5:OFF
	OFF		
6	ON	If SW 8 is set at ON, the communication data length is 32 Bytes. If SW 8 is set at OFF, set SW 6 at OFF.	OFF
	OFF	If SW 8 is set at ON, the communication data length is 17 Bytes. If SW 8 is set at OFF, set SW 6 at OFF.	
7	ON	Software Filter Enable	OFF
	OFF	Software Filter Disable	
8	ON	Communication Speed 10 Mbps	ON
	OFF	Communication Speed 4 Mbps	

(b) Slave Address Setting

Slave Address of Distributed I/O Module must be set using the DIP Switch in a range of 1-5.

Refer to the following table for setting the Slave Address.

Pin No					Slave Address
1	2	3	4	5	
0	0	0	0	0	Not used
1	0	0	0	0	1
0	1	0	0	0	2
1	1	0	0	0	3
0	0	1	0	0	4
1	0	1	0	0	5
0	1	1	0	0	6
1	1	1	0	0	7
0	0	0	1	0	8

1	0	0	1	0	9
0	1	0	1	0	10
1	1	0	1	0	11
0	0	1	1	0	12
1	0	1	1	0	13
0	1	1	1	0	14
1	1	1	1	0	15
0	0	0	0	1	16
1	0	0	0	1	17
0	1	0	0	1	18
1	1	0	0	1	19
0	0	1	0	1	20
1	0	1	0	1	21
0	1	1	0	1	22
1	1	1	0	1	23
0	0	0	1	1	24
1	0	0	1	1	25
0	1	0	1	1	26
1	1	0	1	1	27
0	0	1	1	1	28
1	0	1	1	1	29
0	1	1	1	1	30
1	1	1	1	1	Not used

2.3.2 Performance Specification

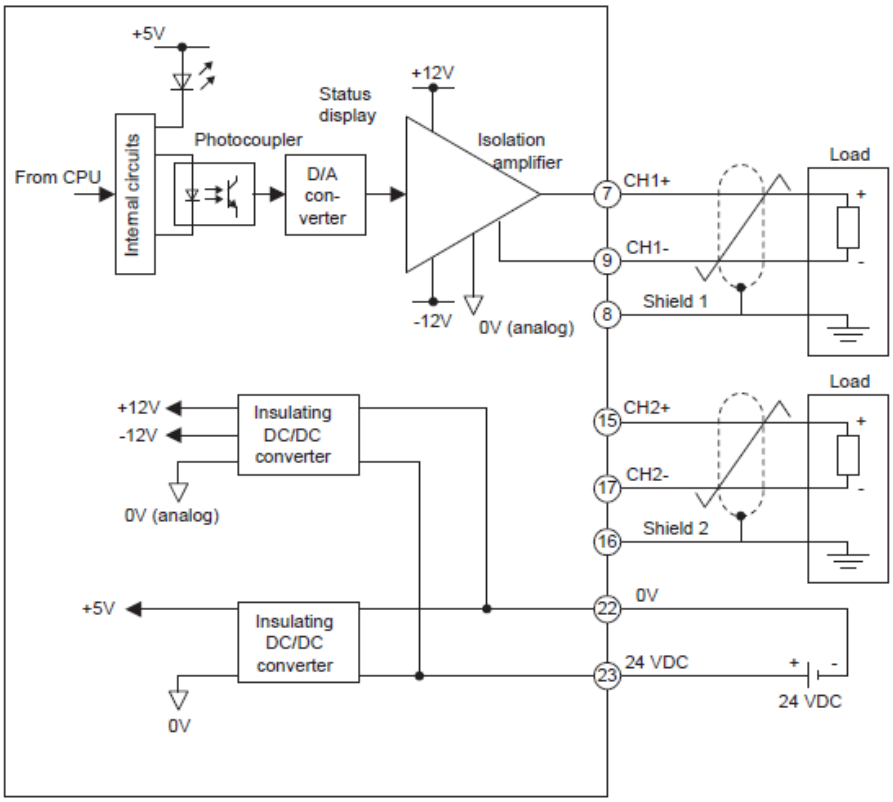
Following Table shows the performance specification of Analog Input Module (±10 V, 4 CH).

Item	Specification
Name	Analog Output Module (±10 V, 2 CH)
Model Name	AN2910
Model Number	JEPMC-AN2910
Input Signal Range	-10 ~ 10 V
Number of Channels	2
Max. Allowable Loading Current	±5mA (2 KΩ)
Digital Resolution	16 bits
Data Form	Binary (2s complement)
Error	±0.2% F.S. (at 25 ℃)

		±0.5% F.S. (at 0 ~ 60 ℃)
Output Delay Time		Max. 4ms
Output Status when disconnected with the Master		SW7 OFF: Output '0'V SW7 ON: Output the previous data from before the communication disconnection
Input Filter		Software Filter
External Connection		Removable Terminal
Status Indication		
Input Circuit Dielectricity	Dielectric Method	Photo-Coupler
	Electric Resistance	DC/DC Insulation
	Dielectric Resistance	100 MΩ
External Input Voltage		24 VDC, 120mA
Dimension		161 x 44 x 79 (W x H x D)

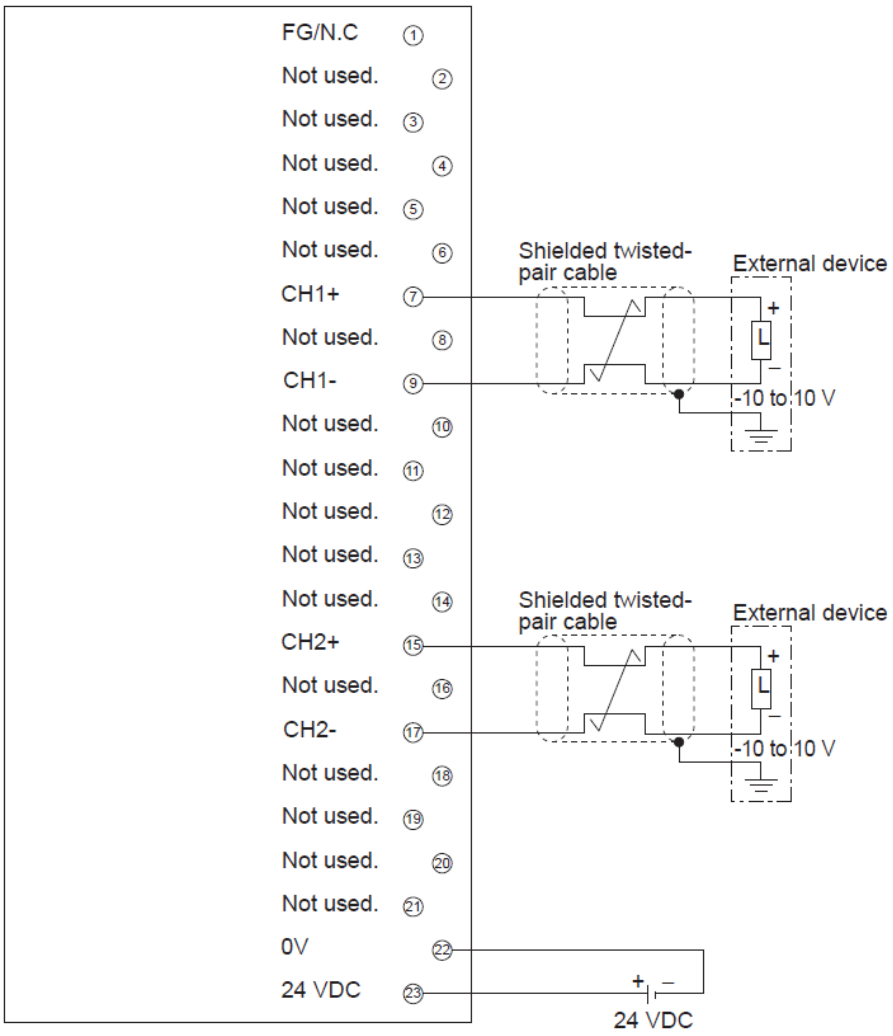
2.3.3 Circuit Configuration

Configure the circuit as shown in the next diagram.



2.3.4 Circuit Wiring Examples

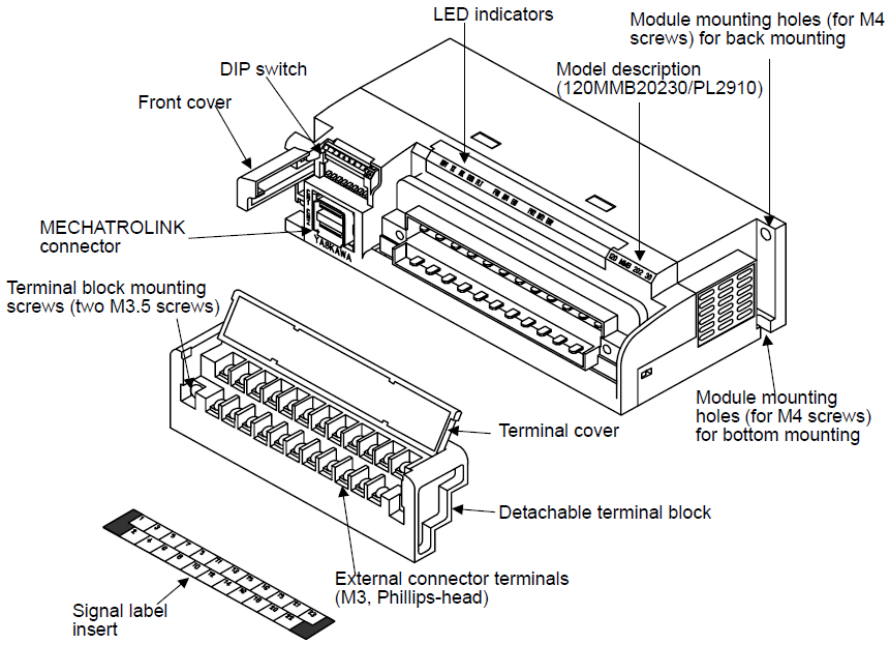
Next diagram shows the examples of terminal block of analog input module.



2.4 Pulse Output Module

2.4.1 Dimension and Configuration

Next diagram shows the dimension and configuration of pulse output module.



(1) LED Indication

RDY	TX	RX	ERR	FLT	FW1	RV1	ER1	FW2	RV2	ER2

Indication	Color	Blinking Meaning
RDY	Green	Normal
TX	Green	Transmitting data
RX	Green	Receiving Data
ERR	Red	Communication Error
FLT	Red	Module Error
FW1	Green	Channel 1 CCW Operating
RV1	Green	Channel 1 CW Operating
ER1	Red	Channel 1 Error
FW2	Green	Channel 2 CCW Operating
RV2	Green	Channel 2 CW Operating
ER2	Red	Channel 2 Error

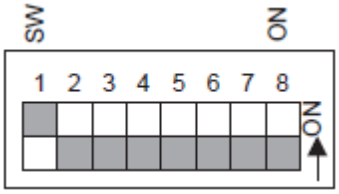
(2) DIP Switch

DIP Switch must be set before operating the Pulse Output Module.

(a) DIP Switch Function

DIP Switch consists of a total of 8 pins.

The pins are numbered from 1 to 8 as shown below. Each pin can move up for turn ON.



Next table shows the functions of each switch.

Pin No	Setting	Function	Factory Setting
1 ~ 5	ON	Set Slave Address	1: ON 2 ~ 5: OFF
	OFF		
6	ON	If SW 8 is set at ON, the communication data length is 32 Bytes. If SW 8 is set at OFF, set SW 6 at OFF.	OFF
	OFF	If SW 8 is set at ON, the communication data length is 17 Bytes. If SW 8 is set at OFF, set SW 6 at OFF.	
7	ON	Software Filter Enable	OFF
	OFF	Software Filter Disable	
8	ON	Communication Speed 10 Mbps	ON
	OFF	Communication Speed 4 Mbps	

(b) Slave Address Setting

Pulse Output Module must be set using the DIP Switch 1-5 for slave address.

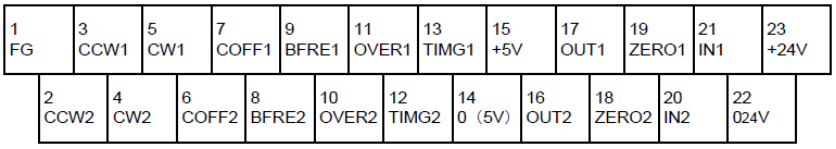
Next table shows the slave address setting.

Pin No					Slave Address
1	2	3	4	5	
0	0	0	0	0	Not used
1	0	0	0	0	1
0	1	0	0	0	2
1	1	0	0	0	3
0	0	1	0	0	4
1	0	1	0	0	5

0	1	1	0	0	6
1	1	1	0	0	7
0	0	0	1	0	8
1	0	0	1	0	9
0	1	0	1	0	10
1	1	0	1	0	11
0	0	1	1	0	12
1	0	1	1	0	13
0	1	1	1	0	14
1	1	1	1	0	15
0	0	0	0	1	16
1	0	0	0	1	17
0	1	0	0	1	18
1	1	0	0	1	19
0	0	1	0	1	20
1	0	1	0	1	21
0	1	1	0	1	22
1	1	1	0	1	23
0	0	0	1	1	24
1	0	0	1	1	25
0	1	0	1	1	26
1	1	0	1	1	27
0	0	1	1	1	28
1	0	1	1	1	29
0	1	1	1	1	30
1	1	1	1	1	Not used

(3) Terminal Block Configuration

Next diagram shows the configuration of terminal block.



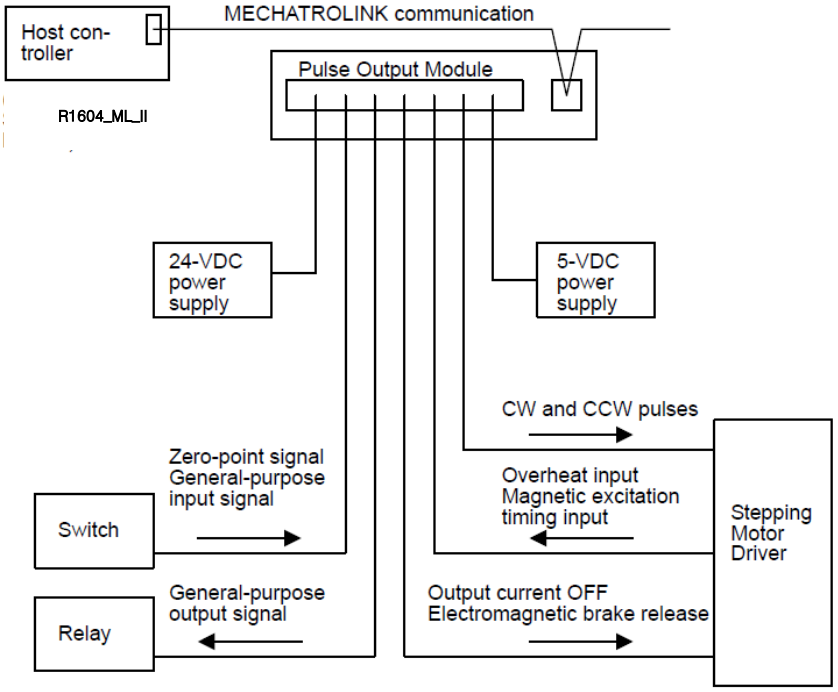
Terminal No.	Signal Name	Signal Function
1	FG	Frame ground
2	CCW2	Channel 2 CCW Pulse Output
3	CCW1	Channel 1 CCW Pulse Output

4	CW2	Channel 2 CW Pulse Output
5	CW1	Channel 1 CW Pulse Output
6	C-OFF2	Channel 2 Output Current OFF
7	C-OFF1	Channel 1 Output Current OFF
8	B-FREE2	Channel2 Break Signal Release
9	B-FREE1	Channel 1 Break Signal Release
10	OVER2	Channel 2 Overheat Input
11	OVER1	Channel 1 Overheat Input
12	TIMING2	Channel 2 Magnetic Excitation Timing Input
13	TIMING1	Channel 1 Magnetic Excitation Timing Input
14	0 (5V)	External Voltage 0V
15	+5V	External Voltage 5V
16	OUT2	Channel 2 all-purpose Output
17	OUT1	Channel 1 all-purpose Output
18	ZERO2	Channel 2 zero signal
19	ZERO1	Channel 1 zero signal
20	IN2	Channel 2 all-purpose input
21	IN1	Channel 1 all-purpose input
22	0 (24V)	External Voltage 0V
23	+24V	External Voltage 5V

2.4.2 System Setting

(1) System Setting Examples

Next diagram shows the examples of System Setting.



2.4.3 Specification

(1) Common Specification

Next table shows the common specification of Output Module.

Item		Specification
Environment Factors	Operating Temperature	0 ~ 55 ℃
	Storage Temperature	-25 ~ 85 ℃
	Operating Humidity	30 ~ 95 %
	Storage Humidity	5 ~ 94 %
Conditions for Equipment Operation	Vibration Resistance	10 ~ 57 Hz with half-amplitude of 0.075mm. 57 ~ 150 Hz at fixed acceleration of 9.8 m/s^2. 10 sweep for 8 minutes each in X,Y,Z directions
	Impact Resistance	Peak acceleration of 147 m/s^2 twice for 11 ms in X,Y,Z directions
Condition for Electric Operation	Noise Resistance	1,000 Vp-p
Installation Condition	Dimension	161 x 44 x 79 (W x H x D)

(2) Performance Specification

(a) Hardware Specification

Next table shows the hardware specification of pulse output module.

Item	Specification
Name	Pulse Output Module (2 Channels)
Model Number	JEPMC – PL2910
Model Name	PL2910
Functions	Pulse Positioning Completion, Jog Function, Original State Recovery
Number of Circuits	2
Communication Protocol	MECHATROLINK II
External Power	24 VDC (20.4 ~ 26.4 VDC)
Internal Consumption Electricity	24V 100mA, 5V 300mA

(b) Performance Specification

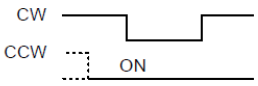
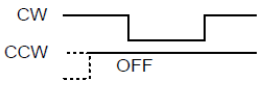
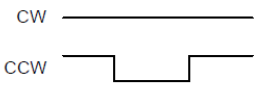
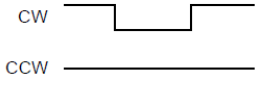
Next table shows the performance specification of pulse output module.

Item		Specification
Pulse Output	Max. Output Speed	500 Kpps
	Pulse Output Voltage	5VDC
	Pulse Input Circuit	Open collector output External input voltage: 5VDC, 10ma/circuit
	External Control Signal	-. Overheat input: 5VDC, sourcing input -. Magnetic excitation timing input: 5VDC, sourcing input -. Output current OFF: 5VDC, sinking output -. Electromagnetic brake release: 5VDC, sinking output -. Zero point signal input: 24VDC, sourcing input -. General-purpose input: 24VDC, sourcing or sinking input -. General-purpose output: 24VDC, sinking input
Detector Function		-. READY: Normal if ON.

	-. PACK: Parameter Setting complete if ON. -. PACK: Parameter Setting incomplete if ON.

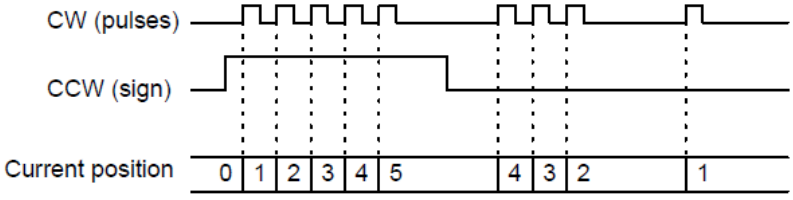
(c) Pulse Output Timing

Next diagram shows the CW and CCW of Pulse Output Timing.

Pulse Output Mode	Forward Timing	Reverse Timing
Sign and Pulses		
CW and CCW pulses		

(d) Pulse Output Action Examples

Next diagram shows examples of "sign and pulses" in Pulse Mode.



(e) External Output Signal Specification

Next table shows the specification of external output signal.

Item		Specification
CW and CCW Pulse Output	Pulse Output Mode	CCW and CW, or sign and pulses
	Max. Output Frequency	500 kpps
	Max. Load Current	18 mA/point
	Output Voltage Drop	0.35 V max.
Output Current in OFF and Output with Electric Brake Released	Rated Voltage	5 VDC
	Range of Allowable Voltage	4.75 ~ 5.25VDC
	Output Form	Sinking
	Max. Load Current	17mA max/point
	Output Voltage Drop	1.0V max
	Output Delay Time	OFF to ON: 1.5ms max

		ON to OFF: 1.5ms max
	Leakage Current at OFF	0.4 μA max
	Output Type	Photo coupler output
All-purpose Outputs	Rated Voltage	24VDC
	Range of Allowable Voltage	19.2 ~ 30VDC
	Output Form	Sinking
	Max. Load Current	100mA max/point
	Output Voltage Drop	1.0V max
	Output Delay Time	OFF to ON: 1.5ms max ON to OFF: 1.5ms max
	Leakage Current at OFF	1mA max

(f) External Input Signal Specification

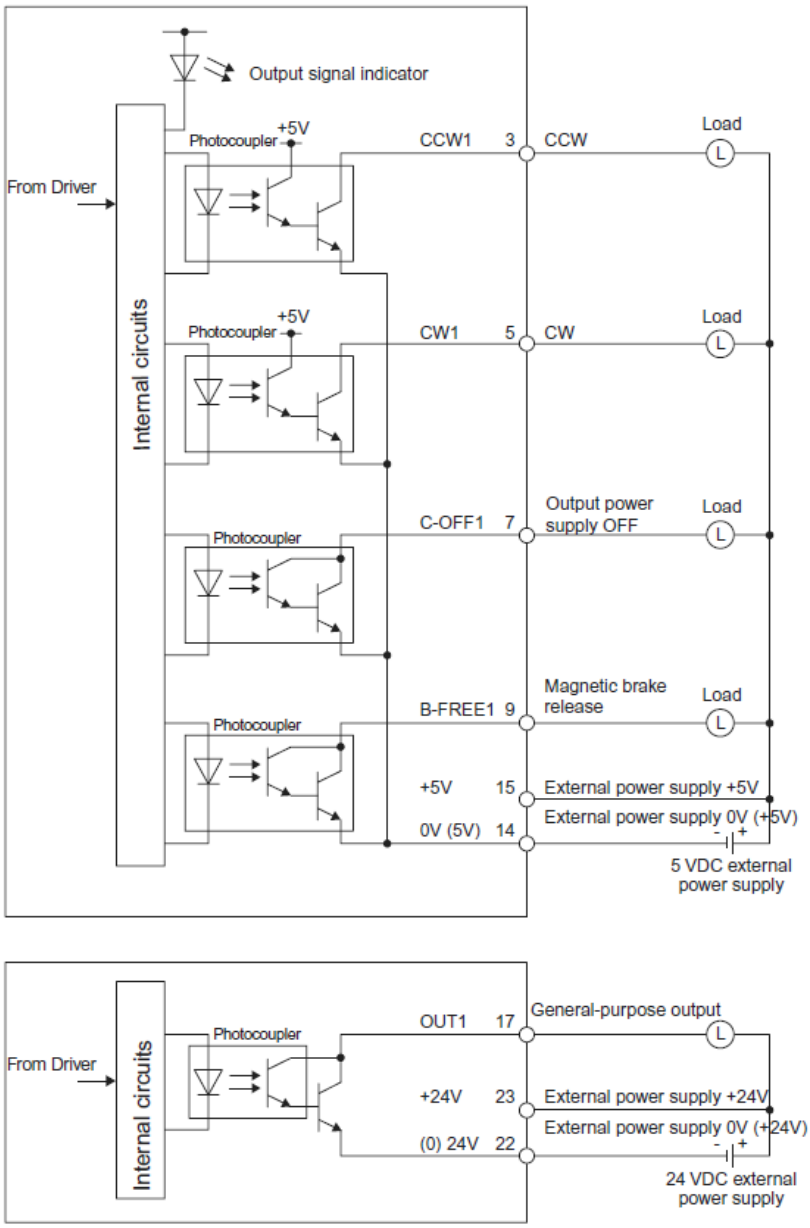
Next table shows the specification of external input signal.

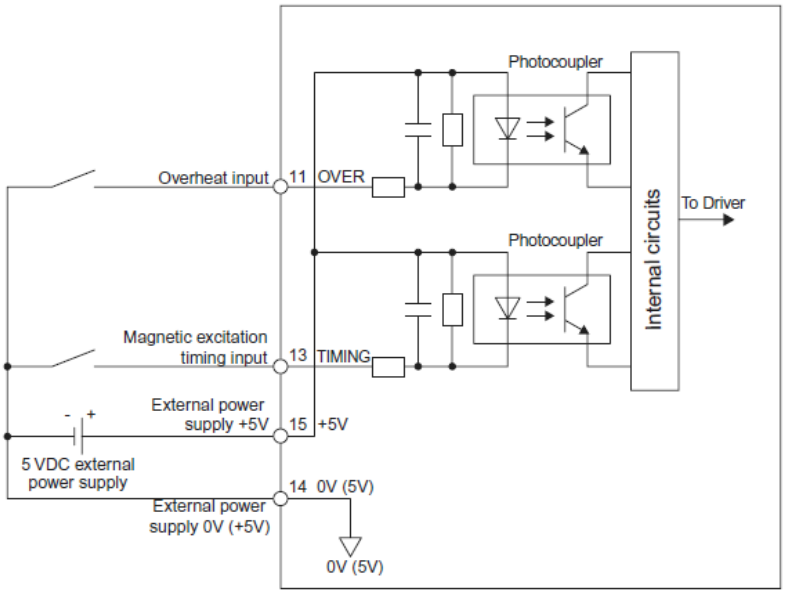
Item		Specification
Overheating and Self-exciting Time	Rated Voltage	5VDC
	Range of Allowable Voltage	4.75 ~ 5.25VDC
	Input Form	Sourcing
	Rated Current	7.0mA
	Input Resistance	470Ω
	Standard Operation Range	Min. ON Voltage: 2.13VDC Max. OFF Voltage: 1.14VDC
	Input Delay Time	OFF to ON: 0.5ms max ON to OFF: 0.8ms max
	Dielectric method	Photo coupler
Original State Signal	Rated Voltage	24VDC
	Range of Allowable Voltage	19.2 ~ 30VDC
	Input Form	Sourcing
	Rated Current	10.0mA
	Input Resistance	2.2KΩ

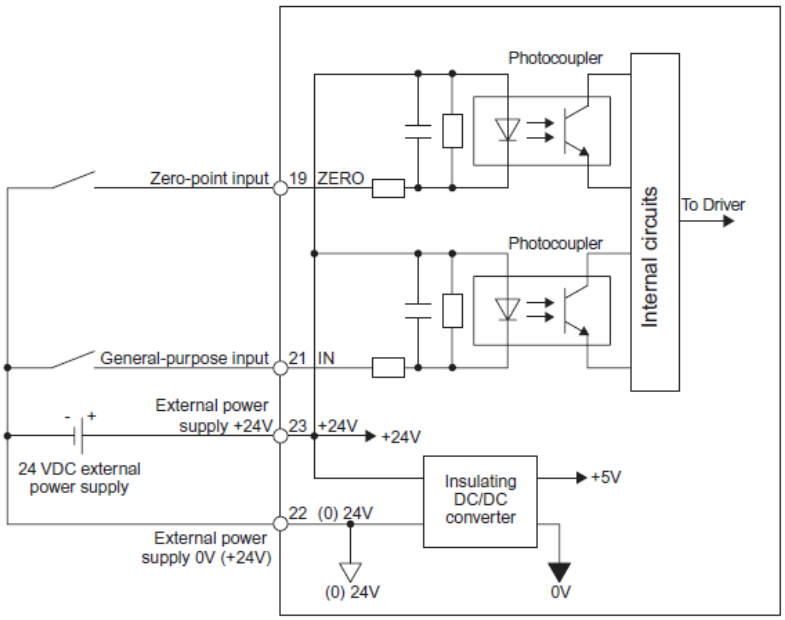
	Standard Operation Range	Min. ON Voltage: 6.9VDC Max. OFF Voltage: 4.8VDC
	Input Delay Time	OFF to ON: 0.5ms max ON to OFF: 0.5ms max
	Dielectric method	Photo coupler
All-purpose Inputs	Rated Voltage	24VDC
	Range of Allowable Voltage	19.2 ~ 30VDC
	Input Form	Sourcing
	Rated Current	5.0mA
	Input Resistance	4.7KΩ
	Standard Operation Range	Min. ON Voltage: 13.1VDC Max. OFF Voltage: 6.3VDC
	Input Delay Time	OFF to ON: 0.5ms max ON to OFF: 0.5ms max
	Dielectric method	Photo coupler

(g) Circuit Configuration

Next diagram shows the circuit configuration of external I/O signals.


Output signal indicator
From Driver
Internal circuits
Photocoupler
+5V
CCW1 3
CCW
Load
CW1 5
CW
C-OFF1 7
Output power supply OFF
B-FREE1 9
Magnetic brake release
+5V 15
External power supply +5V
0V (5V) 14
External power supply 0V (+5V)
5 VDC external power supply
OUT1 17
General-purpose output
+24V 23
External power supply +24V
(0) 24V 22
External power supply 0V (+24V)
24 VDC external power supply





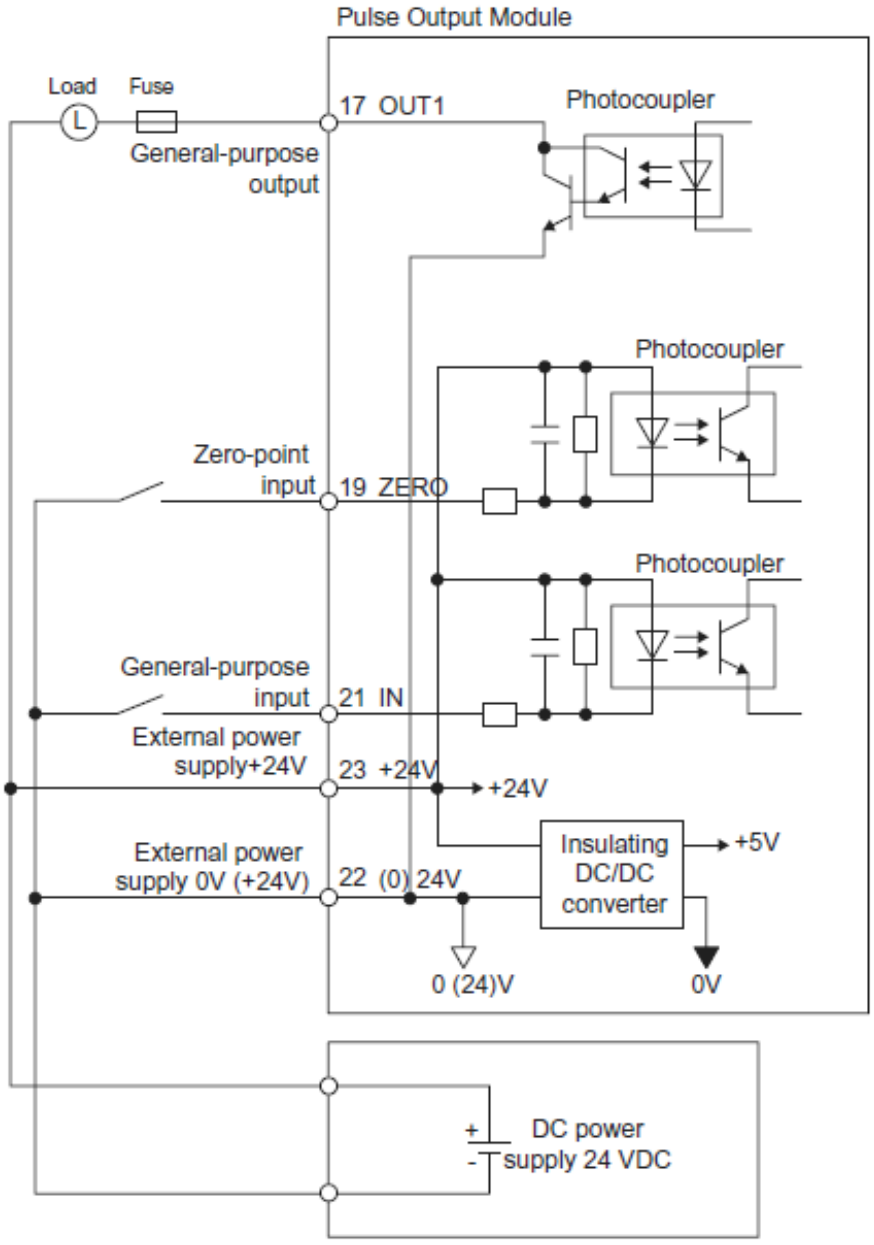
2.4.4 Examples of External I/O Signal and Connection

Next table shows the summary of external I/O signals.

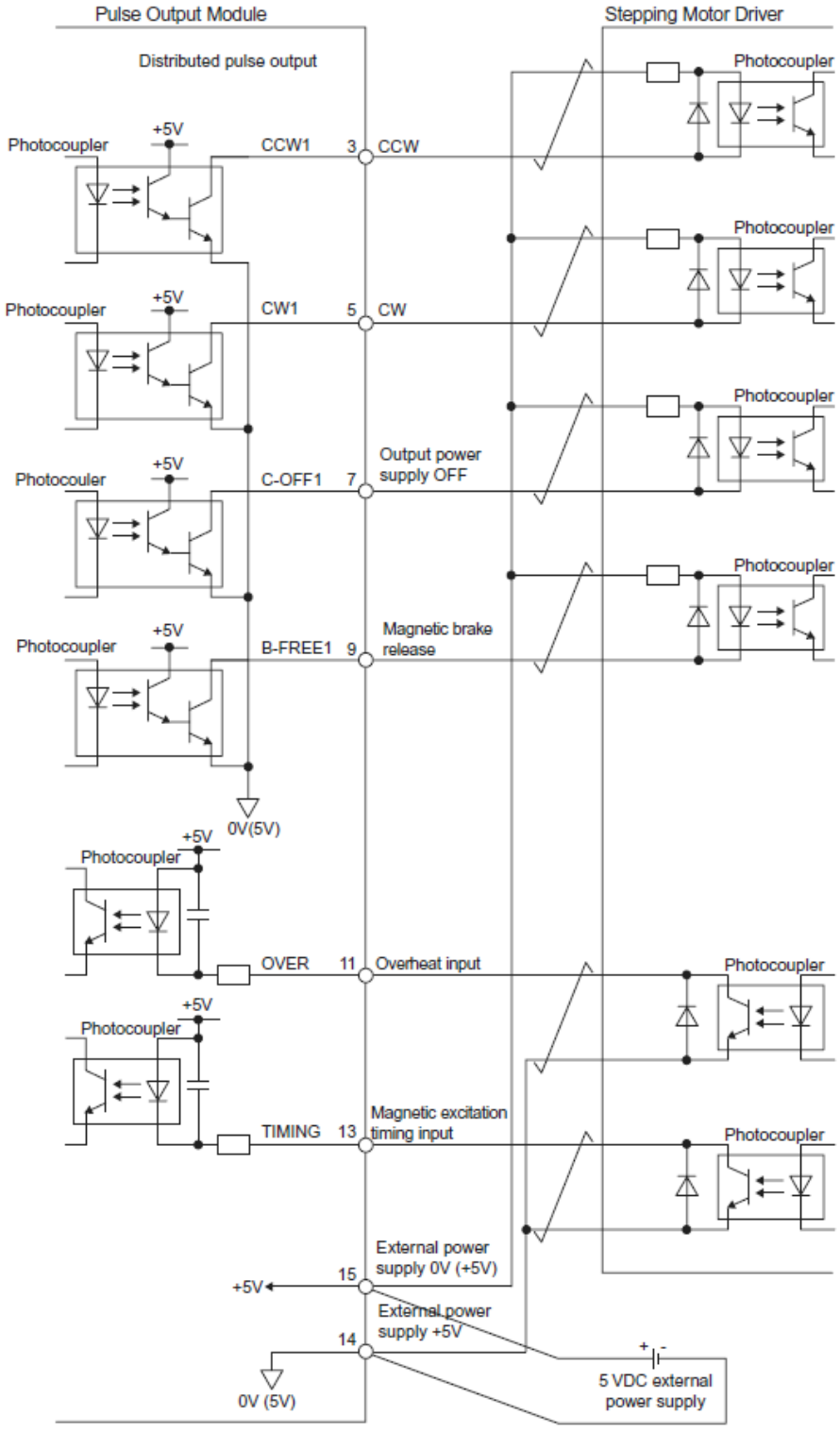
Signal Name		Specification
Output Signals	CWn CCWn	Pulse Output is made from CW or CCW Terminal Pulse Output has 2 methods.
	C-OFFn	Output Current OFF Terminal This signal automatically turns OFF in case of overheating or error in the pulse output module system.
	B-FREEn	Electric brake release output terminal

	OUTn	All-purpose output terminal
Input Signals	OVERn	Output Terminal is connected to the overheating of external device.
	TIMINGn	Signal that indicates self-exciting state.
	ZEROn	Connects to the original state signal.
	INn	Connects to all-purpose input signals.

Diagrams below show examples of all-purpose I/O and original state input signals.



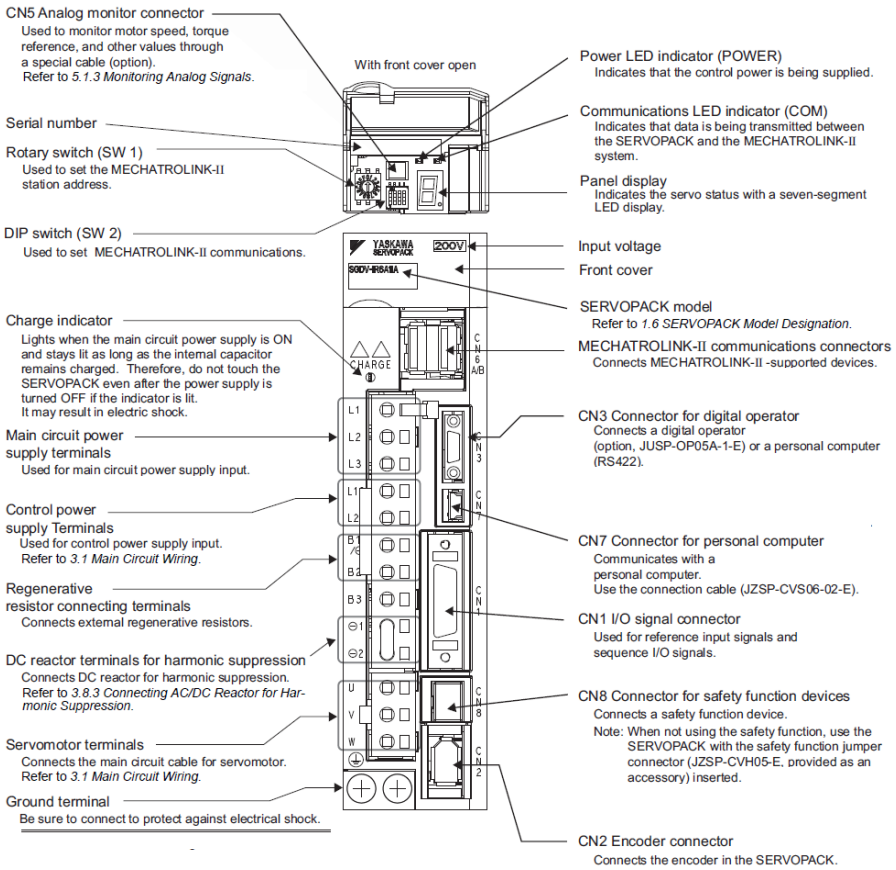
Next diagram shows the examples of stepping motor driver connection.



2.5 SERVOPACK (SGDV-□□□□□□□)

Σ-V Series, a SERVOPACK, has its purpose in increasing its productivity through the maximization of its equipment's potential using high speed/high frequency for accurate location decision.

2.5.1 SERVOPACK Components



2.5.2 SERVOPACK Rating and Specification

Rating and Specification of SERVOPACK are shown below.

(1) Rating

(a) 100 VAC Rating

SGDV (100 VAC)	R70	R90	2R1	2R8
Continuous Output Current [ARMS]	0.66	0.91	2.1	2.8
Max. Output Current	2.1	2.9	6.5	9.3
Main Circuit Power	Single-phase, 100 to 115 VAC +10 ~ -15%, 50/60Hz			
Control Power	Single-phase, 100 to 115 VAC +10 ~ -15%, 50/60Hz			
Overvoltage Class	III			

(b) 200 VAC Rating

SGDV (200 VAC)	R70	R90	1R6	2R8	3R8	5R5	7R6	120	180	200	330	470	550	590	780
Continuous Output	0.	0.	1.	2.	3.	5.	7.	1	1	1	3	4	5	5	7

Current [ARMS]	66	91	6	8	8	5	6	1.6	8.5	9.6	2.9	6.9	4.7	8.6	8.0
Max. Output Current	2.1	2.9	5.8	9.3	11.0	16.9	17	28	42	56	84	110	130	140	170
Main Circuit Power	Three-phase, 200 to 230 VAC +10 ~ -15%, 50/60Hz														
Control Power	Three-phase, 200 to 230 VAC +10 ~ -15%, 50/60Hz														
Overvoltage Class	III														

(c) 400 VAC Rating

SGDV (400 VAC)	1R9	3R5	5R4	8R4	120	170	210	260	280	370
Continuous Output Current [ARMS]	1.9	3.5	5.4	8.4	11.9	16.5	20.8	25.7	28.1	37.2
Max. Output Current	5.5	8.5	14	20	28	42	55	65	70	85
Main Circuit Power	Three-phase, 380 to 480 VAC +10 ~ -15%, 50/60Hz									
Control Power	24VDC ±15%									
Overvoltage Class	III									

(2) Common Specification

Next Table shows the common specification of SERVOPACK.

Control Method		Single-phase or three-phase IGBT-PWM
Feedback		Serial encoder: 13bit(incremental), 17bit, 20bit(incremental/absolute)
Operating Conditions	Operating Temperature/Storage Temperature	Operating Temperature: 0 ~ +55°C, Storage Temperature: –20 ~ 85°C
	Operating Humidity/Storage Humidity	90% RH Below (Stay away from freezing and dew)
	Internal Vibration/Internal Impact	4.9 m/s²/19.8 m/s²
	Protection Class/Pollution Degree	Protection Class: IP1X, Pollution Degree: 2nd Prevent away from combustible and flammable gas, water, oil, and medicine, dust, saline solution, and metallic substances.

	Altitude	1,000m below			
	Other	Prevent away from occurrence of static, noise, strong electric and/or magnetic field, radiation			
Standards		UL508C EN50178, EN55011/A2 group1 class A, EN61000-6-2, EN61800-3, EN61800-5-1, EN954-1, IEC61508-1 to 4			
Configuration		Base-mounted			
Performance		Range of Speed Control	1:5000		
		Speed Fluctuation		Load Fluctuation	Loading: ±0.01% Below (Regular rotational speed)
				Voltage Fluctuation	Rated Voltage ±10%: 0%(Regular rotational speed)
				Temperature Fluctuation	25±25℃ : ±0.1% Below (regular rotational speed)
		Torque Control Limit (Reproducibility)	±1%		
		Soft Start Time Setting	0 ~ 10s(Acceleration・Deceleration Setting Available)		
Input/Output Signal		Encoder Divided Pulse Output	A, B & C: Line Driver Output Number of Divided Pulses: Any Setting Possible		

	Sequence Input Signal	Assignable Input Signal	Number of Channels	7
			Function	Homing Deceleration Switch Signal (/DEC), External Latch Signal (/EXT1-3), Forward Run Prohibited (P-OT), Reverse Run Prohibited (N-OT), Forward External Torque Limit (/P-CL), Reverse External Torque Limit (/N-CL) Changes to the signals and forward/reverse logic possible.
	Sequence Output Signal	Fixed Output	Servo Alarm (ALM)	
		Assignable Output Signal	Number of Channels	3
			Function	Positioning Completion (/COIN), Speed Coincidence Detection (/V-CMP), Rotation Detection (/TGON), Servo Ready (/S-RDY), Torque Limit Detection (/CLT), Speed Limit

				Detection (/VLT), Brake (/BK), Warning (/WARN), Near (/NEAR) Changes to the signals and forward/reverse logic possible.
Communication Function	RS422A Communication (CN3)	Connection Equipment	Digital Operator (JUSP-OP05A-1-E) , PC (SigmaWin+Response)	
		1:N Communication	N=15 Possible for RS422A Port	
		Axis Address Setting	Settings must be made with Parameter concerned	
	USD Communication (CN7)	Connection Equipment	PC(SigmaWin + Response)	
		Communication Standards	USB1.1 Standard (12 Mbps)	
Indication Function	CHARGE, 7 segments LED × 5 locations			
Analog Monitor Unit Function (CN5)	Number of Terminals: 2 Range of Voltage Output DC ±10 V(Linearity Effective Range ±8 V) Resolution : 16 bit Limit±20 mV (Typ) Max. Output Current±10 mA Settling Time (±1%) : 1.2 ms (Typ)			
Dynamic Brake (DB)	Operate if main power OFF, servo alarm, servo off, overtravelling (OT)			
Regenerative Processing	International Regenerative Resistor or External Regenerative			

	Resistor (Option)	
Overtravelling Prevention	Deceleration or Free-run STOP for Dynamic Brake (DB) when operating P-OT, N-OT Inputs	
Protection Function	Overcurrent, Overvoltage, Undervoltage, overloading, Regeneration	
Support Function	Gain Adjustment, Alarm Record, Jog Operation, Original State Search	
Safety Function	Input	/HWBB1, /HWBB2: Base Block Signal of Power Module
	Output	EDM1: Internal Safety Circuit Status Detection (Fixed Output)
Option Card	Full Close Option Card	

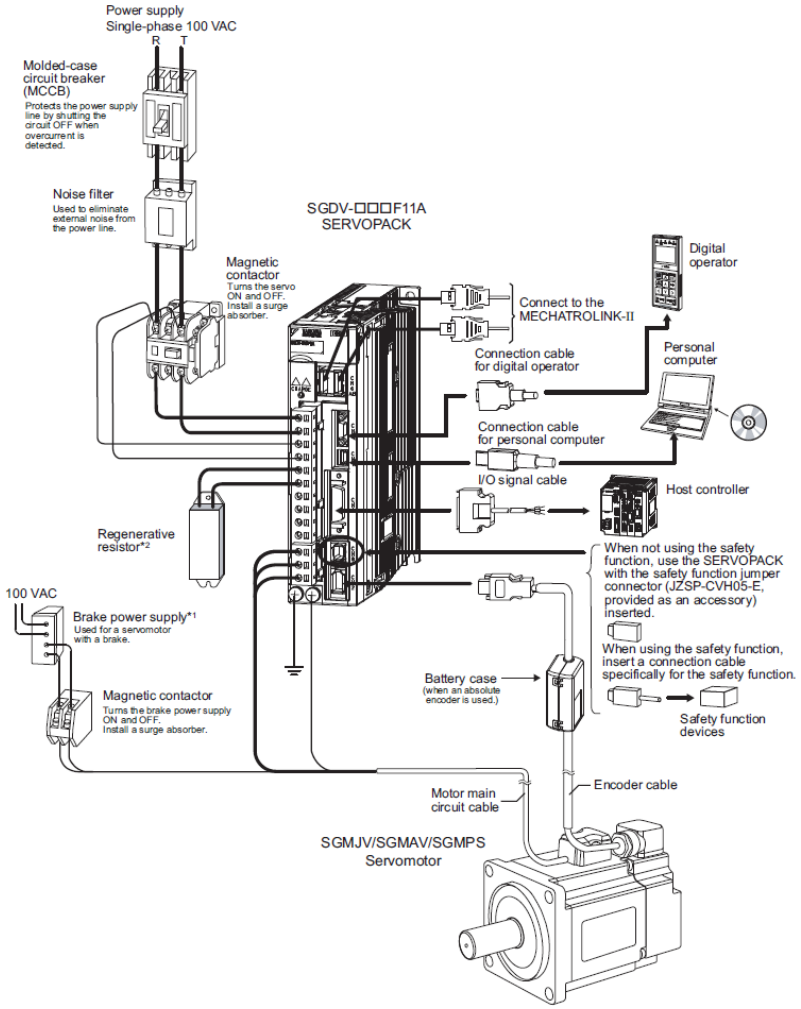
(3) MECHATROLINK II Function Specification

Next table shows the function specification of MECHATROLINK II.

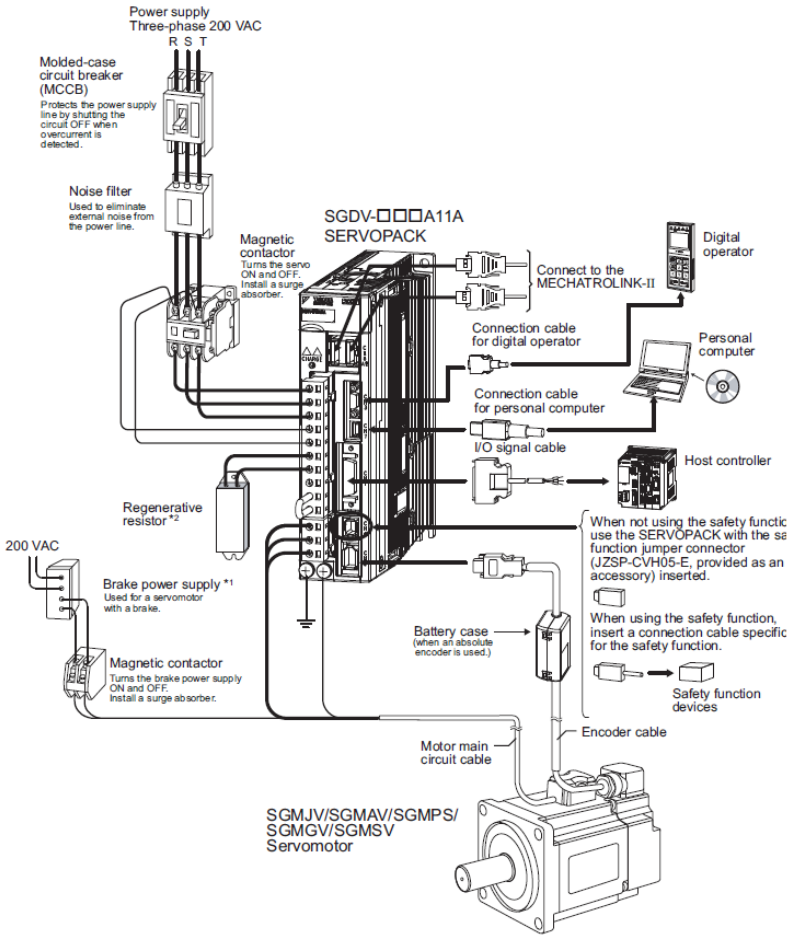
Function		Specification
MECHATROLINK-II Communication	Communication Rule	MECHATROLINK-II
	Station Address	41H ~ 5FH
	Baud Rate	10 Mbps
	Communication Cycle	1ms
	Communication Data Size	17Bytes
Standard Methods	Control Method	Positioning, Speed, and Torque Control for MECHATROLINK-II Communication
	Standard Input	MECHATROLINK-II Commands

2.5.3 Examples of Servo System Configuration

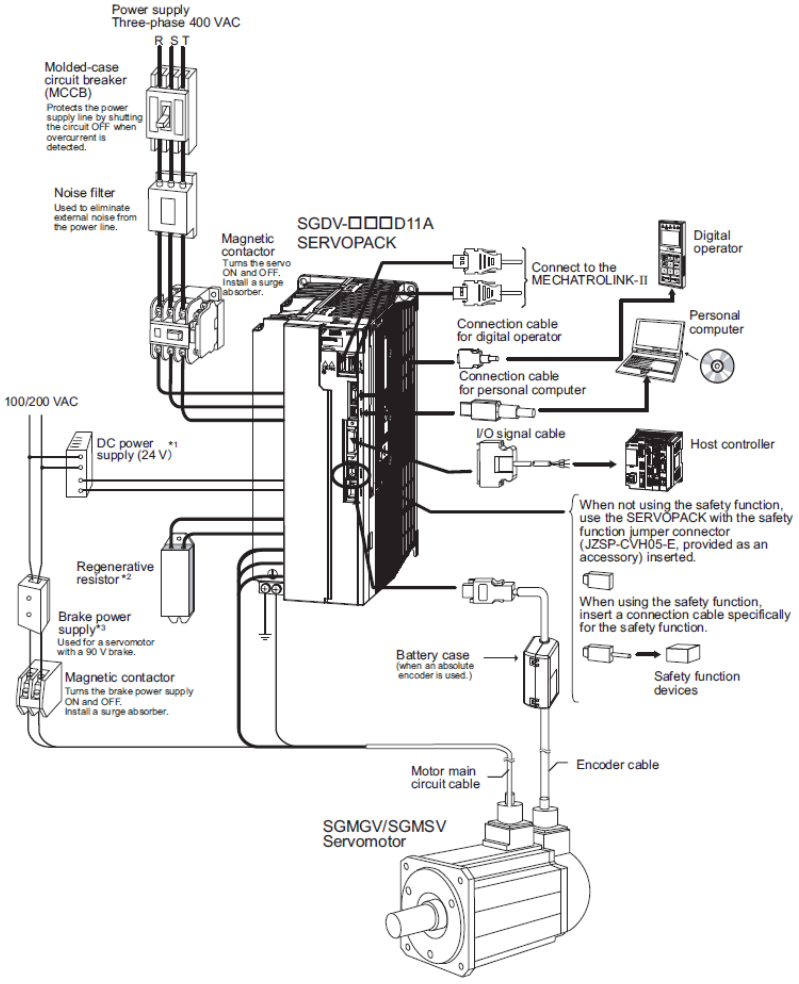
(1) Single-Phase 100 VAC



(2) Three-Phase 200 VAC

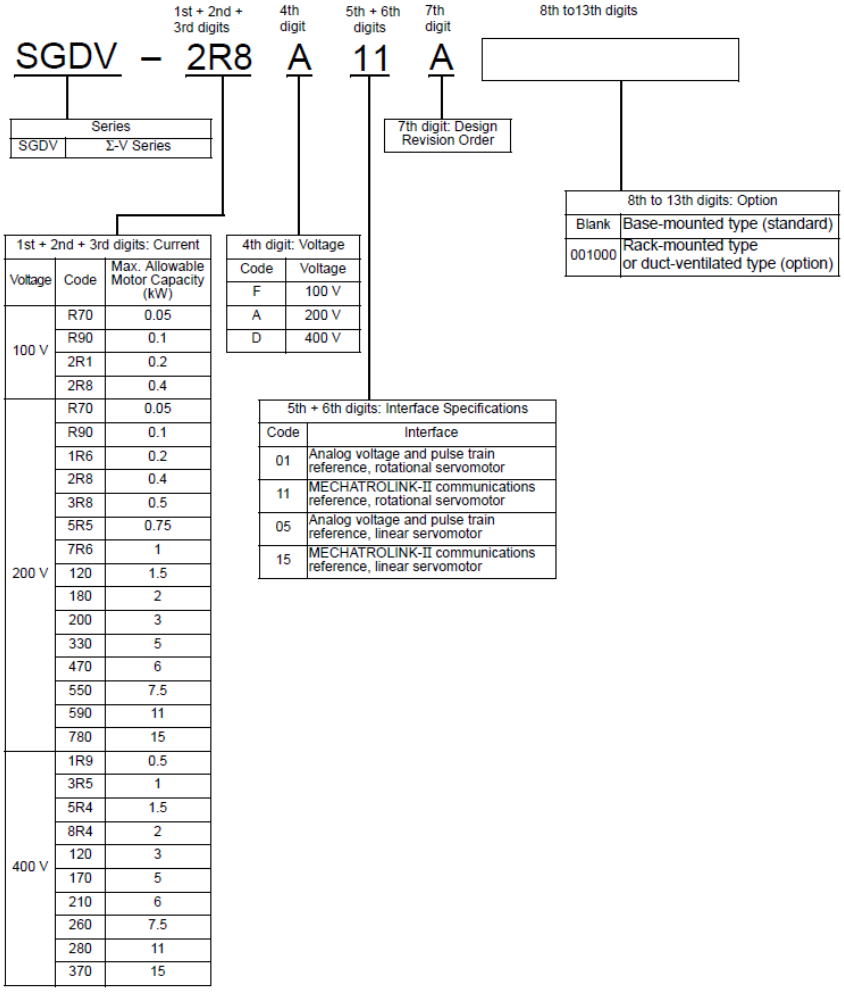


(3) Three-Phase 400 VAC



2.5.4 SERVOPACK Model Names

Select the SERVOPACK accordingly to the selection made for Servo Motor.



2.5.5 Panel Display

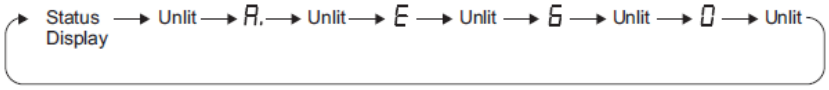
Gauges of SERVOPACK show the status.

(1) Status View

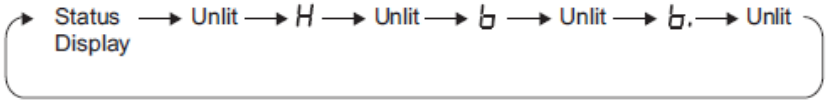
Display	Meaning
	Rotation Detection (/TGON) Lights if motor speed exceeds the value set in Pn502. (Factory setting: 20 min-1)
	Base-block Lights for base-block. Does not light when servomotor power is ON.
	Reference Input Lights when a reference is being input.
	CONNECT Lights during connection.

(2) Alarm and Warning Display

Example: Alarm A.E60

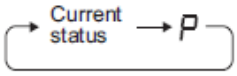


(3) Hard Wire Base Block Display

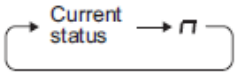


(4) Over travel Display

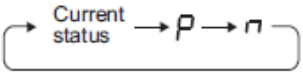
① Overtravel at forward rotation (P-OT)



② Overtravel at reverse rotation (N-OT)

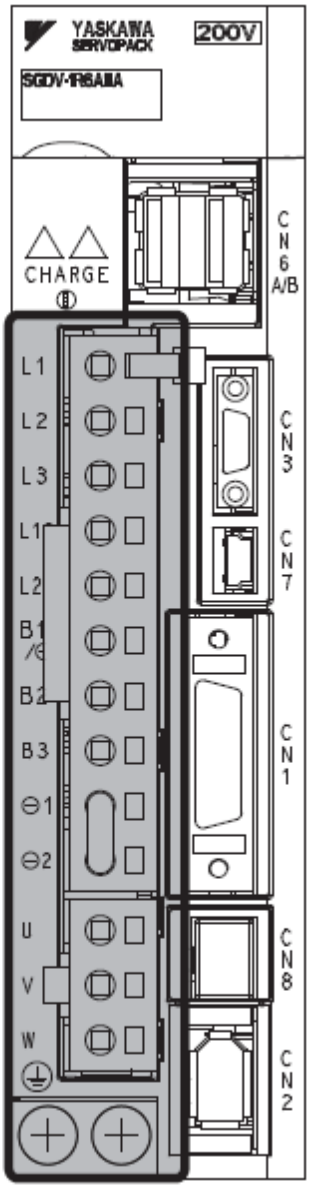


③ Overtravel at forward/reverse rotation



2.5.6 Wiring and Connection

(1) Main Circuit Wiring



: Main terminals

Name	Terminal Symbols	Model SGDV-□□□□	Information
Main Circuit Input Terminal	L1, L2	□□□F	Single-phase 100 to 115 V, +10% to -15% (50/60 Hz)
	L1, L2, L3	□□□A	Three-phase 200 to 230 V, +10% to -15% (50/60 Hz)
		□□□D	Three-phase 380 to 480 V, +10% to -15% (50/60 Hz)

Control Power Input Terminal	L1C, L2C	□□□F	Single-phase 100 to 115 V, +10% to -15% (50/60 Hz)
		□□□A	Single-phase 200 to 230 V, +10% to -15% (50/60 Hz)
	24V, 0V	□□□D	24 VDC, ±15%
External Regenerative Resistor Terminal	B1/⊕, B2 or B1, B2	R70F, R90F, 2R1F, 2R8F, R70A, R90A, 1R6A, 2R8A	Connect the external regenerative resistor (option) in between B1/-B2 in case of low regenerative ability
		3R8A, 5R5A, 7R6A, 120A, 180A, 200A, 330A, 1R9D, 3R5D, 5R4D, 8R4D, 120D, 170D	Open (wiring separated) between B2 and B3*1, and connect the external regenerative resistor between B1/⊕ -B2 in case of Small Capacity with internal regenerative resistor
		470A, 550A, 590A, 780A, 210D, 260D, 280D, 370D	Connect the Regenerative Resistor between either B1/ ⊕ and B2 or B1 and B2
Power high-frequency suppression DC reactor terminal	⊖1, ⊖2	□□□A □□□D	Connect the DC Reactor in between ⊖1 - ⊖2*2 in case the Power High frequency requires suppression
Main circuit plus terminal	B1/ ⊕or B1	□□□A □□□D	Use in case of DC Power input.
Main circuit minus terminal	⊖ 2 or ⊖	□□□A □□□D	
Servomotor Connection Terminal	U, V, W	Connect with servo motor.	
Ground terminals (x2)	⏚	Connect the power and motor grounding terminals together for grounding terminal.	

(2) SERVOPACK Main Circuit Line and Size

(a) Line Form

Line Form		Conductor Ambient Temperature(℃)
Symbol	Name	

IV	600V PVC insulated wire	60
HIV	Special PVC insulated wire	75

(b) Single-Phase 100V

External Terminal Name	Terminal Symbol	SERVOPACK Model SGDV-			
		R70	R90	2R1	2R8
Main Circuit Power Input Terminal	L1 , L2	HIV1.25		HIV2.0	
Control Power Input Terminal	L1C , L2C	HIV1.25			
Servo motor Connection Terminal	U , V , W	HIV1.25			
External Regenerative Resistor Connection Terminal	B1/ ⊕ , B2	HIV1.25			
Grounding Terminal	⏚	HIV2.0 or higher			

(c) Three-phase 2000V

External Terminal Name	Terminal Symbol	SERVOPACK Model SGDV-														
		R70	R90	1R6	2R8	3R8	5R5	7R6	120	180	200	330	470	550	590	780
Main Circuit Power Input Terminal	L1, L2, L3	HIV1.25			HIV2.0					HIV 3.5		HIV5.5	HIV8.0	HIV14.0		HIV2 2.0
Control Power Input Terminal	L1C, L2C	HIV1.25														
Servo motor Connection Terminal	U, V, W	HIV1.25				HIV2.0				HIV 3.5	HIV 5.5	HIV 8.0		HIV 14.0		HIV2 2.0
External Regenerative Resistor Connection Terminal	B1/⊕, B2	HIV1.25								HIV 2	HIV 3	HIV 5		HIV 8.0		HIV2 2.0

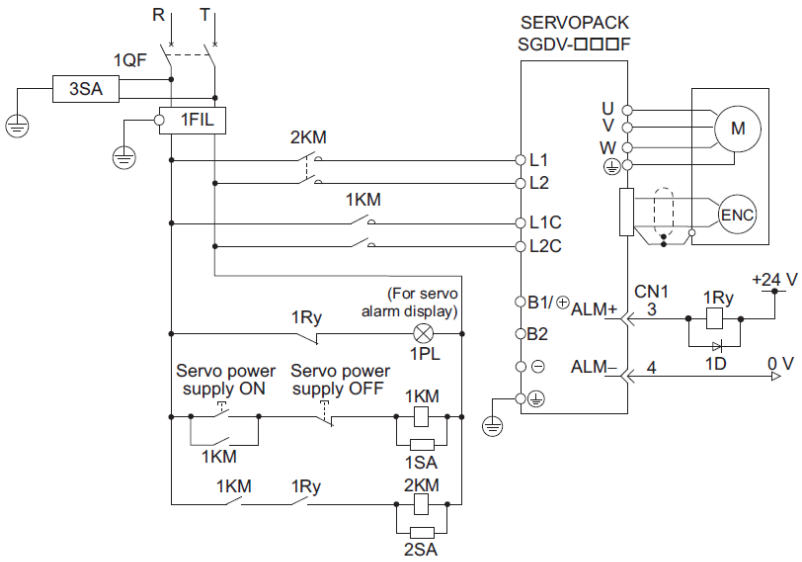
			.0	.5	.5		
Grounding Terminal	⏚	HIV2.0 or higher					

(d) Three-phase 400V

External Terminal Name	Terminal Symbol	SERVOPACK Model SGDV-									
		1R9	3R5	5R4	8R4	120	170	210	260	280	370
Main Circuit Power Input Terminal	L1, L2, L3	HIV1.25			HIV2.0		HIV3.5		HIV 5.5	HIV 8.0	HIV 14.0
Control Power Input Terminal	24 V, 0 V	HIV1.25									
Servo motor Connection Terminal	U, V, W	HIV1.25			HIV2.0		HIV 3.5	HIV5.5		HIV 8.0	HIV 14.0
External Regenerative Resistor Connection Terminal	B1/ ⊕ , B2	HIV1.25					HIV2.0	HIV3.5		HIV 5.5	HIV 8.0
Grounding Terminal	⏚	HIV2.0 or higher									

(3) Main Circuit Runoff Examples

(e) Single-phase 100 V, SGDV-□□□F (SGDV-R70F, R90F, 2R1F, 2R8F)

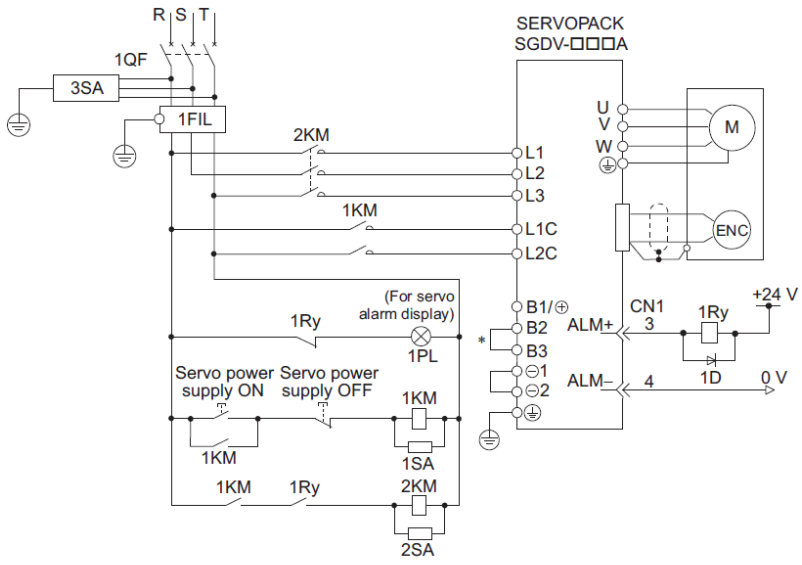


1QF: Molded-case circuit breaker
1FIL: Noise filter
1KM: Magnetic contactor (for control power supply)
2KM: Magnetic contactor (for main power supply)
1Ry: Relay

1PL: Indicator lamp
1SA: Surge absorber
2SA: Surge absorber
3SA: Surge absorber
1D: Flywheel diode

(f) Three-phase 200 V, SGDV-□□□A

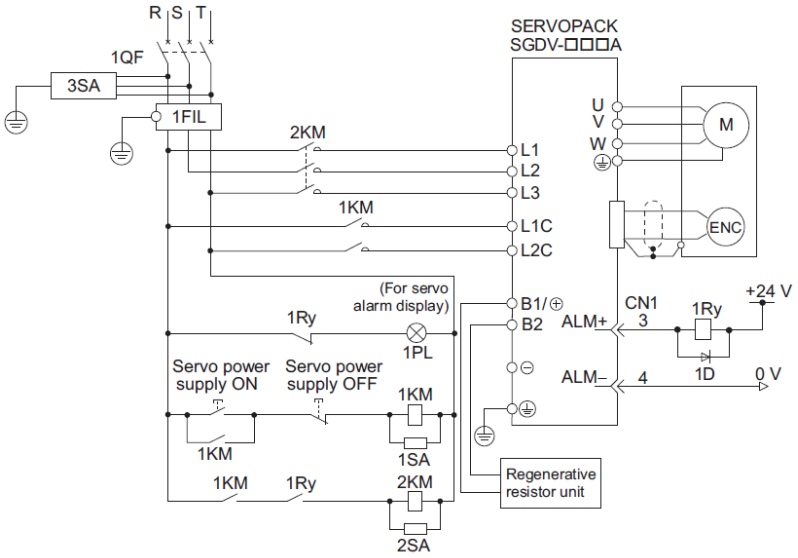
-. SGDV-R70A, R90A, 1R6A, 2R8A, 3R8A, 5R5A, 7R6A, 120A, 180A, 200A, 330A)



1QF: Molded-case circuit breaker
1FIL: Noise filter
1KM: Magnetic contactor (for control power supply)
2KM: Magnetic contactor (for main power supply)
1Ry: Relay

1PL: Indicator lamp
1SA: Surge absorber
2SA: Surge absorber
3SA: Surge absorber
1D: Flywheel diode

-. SGDV-470A, 550A, 590A, 780A

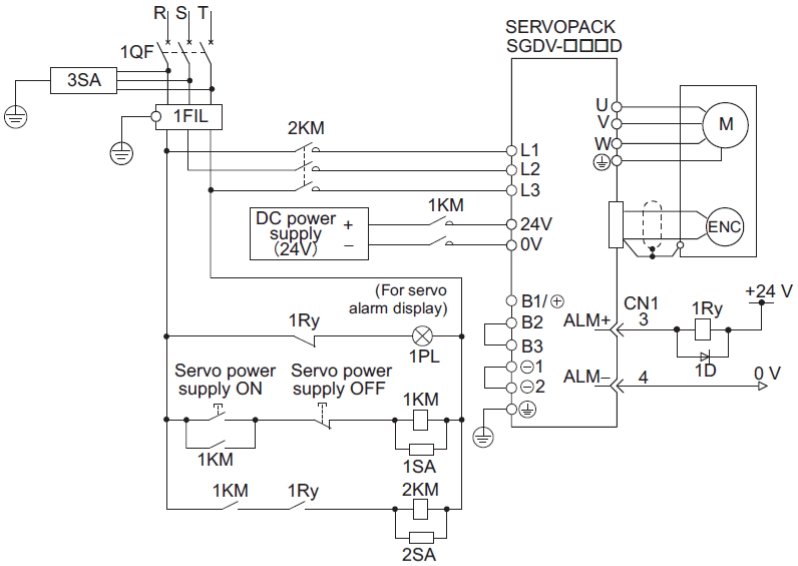


1QF: Molded-case circuit breaker
1FIL: Noise filter
1KM: Magnetic contactor (for control power supply)
2KM: Magnetic contactor (for main power supply)
1Ry: Relay

1PL: Indicator lamp
1SA: Surge absorber
2SA: Surge absorber
3SA: Surge absorber
1D: Flywheel diode

(g) Three-phase 400 V, SGDV-□□□D

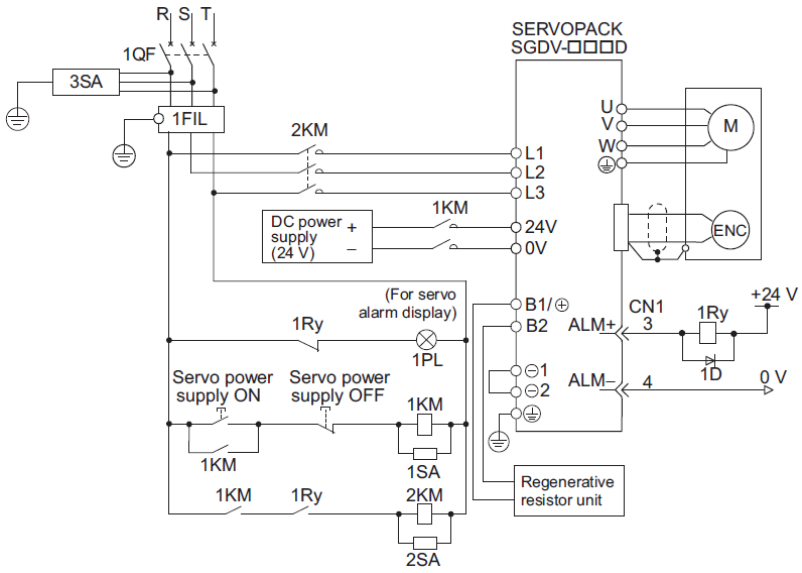
-. SGDV-1R9D, 3R5D, 5R4D, 8R4D, 120D, 170D



1QF: Molded-case circuit breaker
1FIL: Noise filter
1KM: Magnetic contactor (for control power supply)
2KM: Magnetic contactor (for main power supply)
1Ry: Relay

1PL: Indicator lamp
1SA: Surge absorber
2SA: Surge absorber
3SA: Surge absorber
1D: Flywheel diode

-. SGDV-210D, 260D, 280D, 370D



1QF: Molded-case circuit breaker
1FIL: Noise filter
1KM: Magnetic contactor (for control power supply)
2KM: Magnetic contactor (for main power supply)
1Ry: Relay
1PL: Indicator lamp
1SA: Surge absorber
2SA: Surge absorber
3SA: Surge absorber
1D: Flywheel diode

(4) Caution in case of using SERVOPACK with DC Power Input

Change Pn001.2 to 「1」 if the DC Power Input is used for SERVOPACK

(h) Main Circuit/Control Power Input

-. Three-phase 200V

SERVOPACK Model SGDV-	Terminal Names and Information		
	Main Circuit Plus Terminal	Main Circuit Minus Terminal	Control Power Input Terminal
	270 ~ 320 VDC	0 VDC	200 ~ 230 VAC
R70A , R90A , 1R6A , 2R8A , 3R8A , 5R5A , 7R6A , 120A , 180A , 200A , 330A	B1/⊕	⊖2	L1C, L2C
470A, 550A, 590A, 780A	B1/⊕	⊖	L1C, L2C

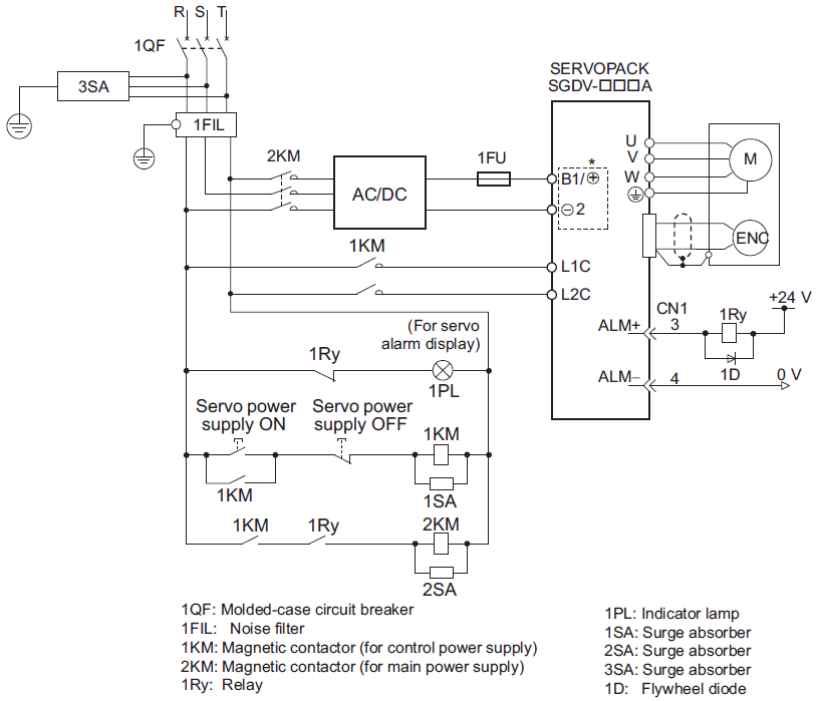
-. Three-phase 400V

SERVOPACK Model SGDV-	Terminal Names and Information		
	Main Circuit Plus Terminal	Main Circuit Plus Terminal	Main Circuit Plus Terminal
	513 ~ 648 VDC	0 VDC	24VDC (± 15%)

1R9D, 3R5D, 5R4D, 8R4D, 120D, 210D, 260D, 280D, 370D	B1/⊕	⊖2	24 V, 0 V
170D	B1/⊕	⊖	24 V, 0 V

(i) DC Power Input Runoff Examples

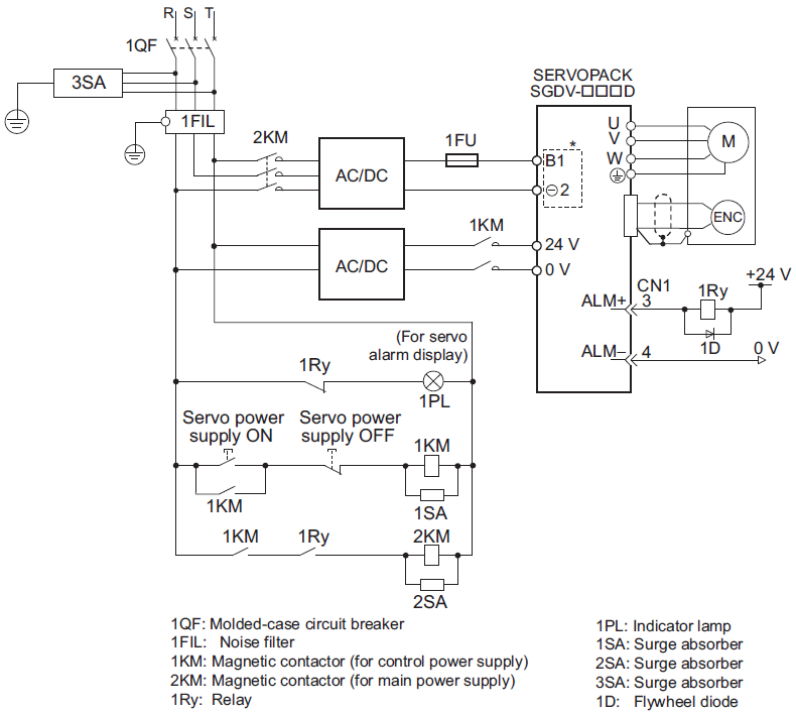
-. 200 V SERVOPACK SGDV-□□□A



1QF: Molded-case circuit breaker
1FIL: Noise filter
1KM: Magnetic contactor (for control power supply)
2KM: Magnetic contactor (for main power supply)
1Ry: Relay

1PL: Indicator lamp
1SA: Surge absorber
2SA: Surge absorber
3SA: Surge absorber
1D: Flywheel diode

-. 400 V SERVOPACK SGDV-□□□D



-. Parameter Setting

Change Parameter Pn001.2 to 「1」 if the DC Power Input is used for SERVOPACK

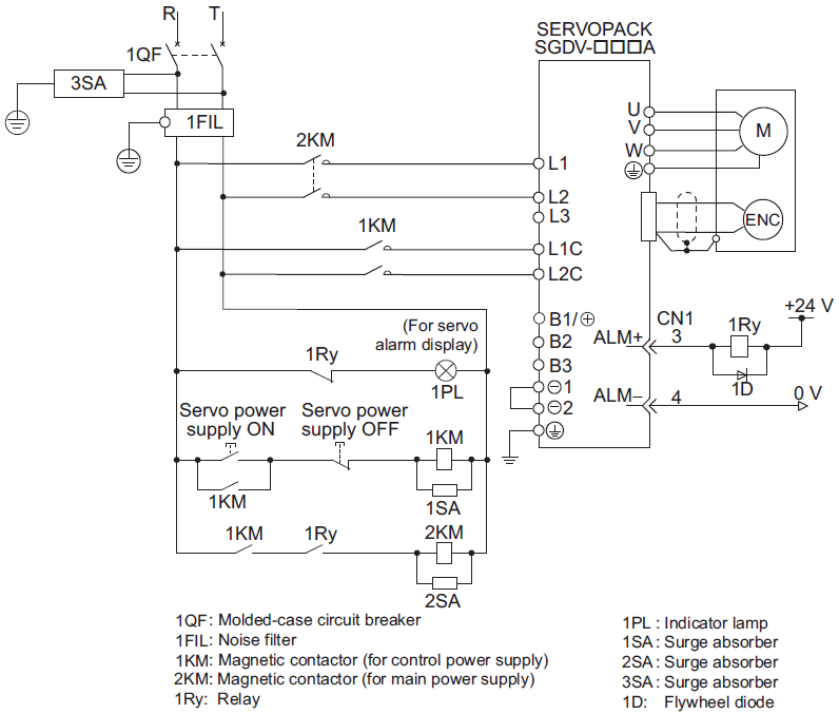
Parameter		Meaning	Applicable Point	Type
Pn001	n.□0□□	Respond to AC Power Input	After re-insertion of Power	Setting
	n.□1□□	Respond to DC Power Input		

(5) Caution in case the single-phase 200V power input uses SERVOPACK

In the Σ-V Series, there are single and three phase 200Vs that support SERVOPACK. (SGDV-R70A, R90A, 1R6A, 2R8A, 5R5A)

Parameter		Meaning	Applicable Point	Type
Pn00B	n.□0□□	There-phase power Enable	After re-insertion of power	Setting
	n.□1□□	Single-phase power Enable		

Single-phase 200V SERVOPACK SGDV-R70A, R90A, 1R6A, 2R8A, 5R5A Runoff Example



(6) I/O Signal Connection

(a) Input Signal

Signal	Pin No.	Name	Function
/DEC	9	Homing Deceleration Limit Switch	Do not connect
P-OT N-OT	7 8	CW Rotation Prohibited CCW Rotation Prohibited	Do not connect
/EXT1 /EXT2 /EXT3	10 11 12	External Latch Signal 1 External Latch Signal 2 External Latch Signal 3	Home Sensor Input + End Limit signal Connection - End limit signal connection
+24VIN	6	Control Power	Control Power
BAT(+) BAT(-)	14 15	Battery (+) Input Battery (-) Input	Absolute encoder back-up battery connection pin
/SIO	13	All-purpose input	All-purpose signal

(b) Output Signal

Signal	Pin No.	Name	Function
ALM+ ALM-	3 4	Servo Alarm Output	Turns OFF if error occurs
/BK+ (/SO1+)	1	Brake Signal	Brake Control

/BK- (/SO-)	2		
/SO2+ /SO2- /SO3+ /SO3-	23 24 25 26	All-purpose output	All-purpose output
FG	Shell	Frame Ground	Connect Frame Ground

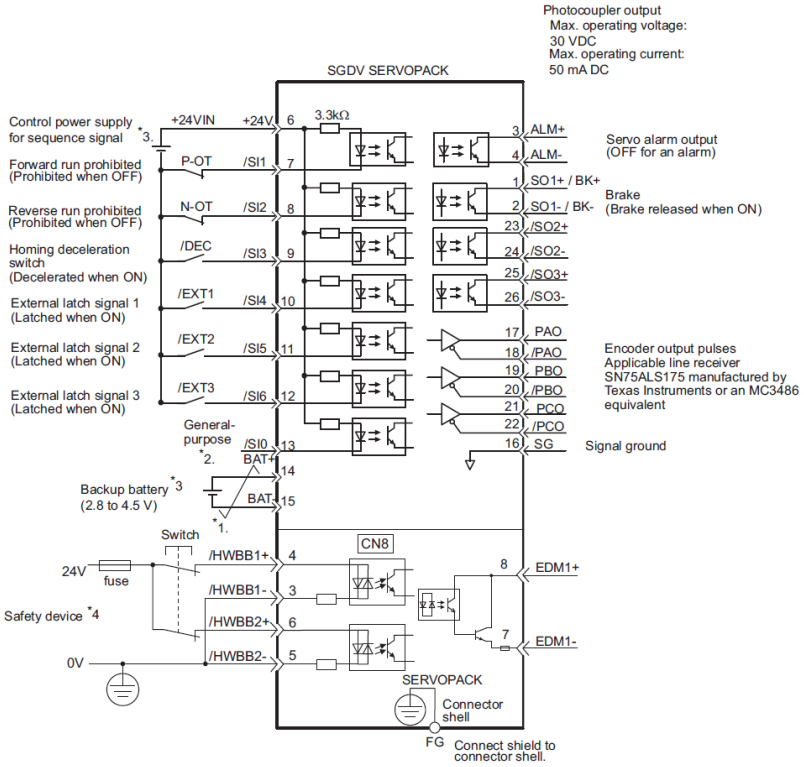
(c) I/O Signal Connection (CN1) Terminal Configuration

1	/BK+ (/SO1+)	Brake output	2	/BK- (/SO1-)	Brake output	14	BAT(+)	Battery (+) input	15	BAT(-)	Battery (-) input
3	ALM+	Servo alarm output	4	ALM-	Servo alarm output	16	SG	Signal ground	17	PAO	Encoder output pulse Phase A
5			6	+24VIN	Control power supply for sequence signal input	18	/PAO	Encoder output pulse Phase A	19	PBO	Encoder output pulse Phase B
7	P-OT (/SI1)	Forward run prohibited input	8	N-OT (/SI2)	Reverse run prohibited input	20	/PBO	Encoder output pulse Phase B	21	PCO	Encoder output pulse Phase C
9	/DEC (/SI3)	Zero-point return deceleration switch input	10	/EXT1 (/SI4)	External latch signal 1 input	22	/PCO	Encoder output pulse Phase C	23	/SO2+	General-purpose input
11	/EXT2 (/SI5)	External latch signal 2 input	12	/EXT3 (/SI6)	External latch signal 3 input	24	/SO2-	General-purpose input	25	/SO3+	General-purpose input
13	/SI0	General-purpose input				26	/SO3-	General-purpose input			

(d) Safety Function Sign (CN8) Names and Function

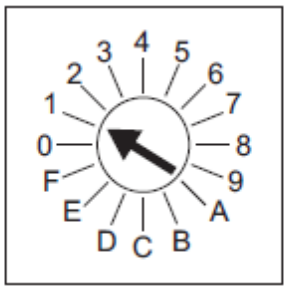
Pin No,	Signal Names	Function	
1	-	-	-
2	-	-	-
3	/HWBB1-	Hard wire base block Input 1	Base Block (Motor current prohibited) occurs if Signal OFF
4	/HWBB1+		
5	/HWBB2-	Hard wire base block Input 2	
6	/HWBB2+		
7	EDM1-	Detection Circuit Status for Output	Status turns ON if the Hardwire base block function is operational
8	EDM1+		

(e) I/O Signal Connection Example

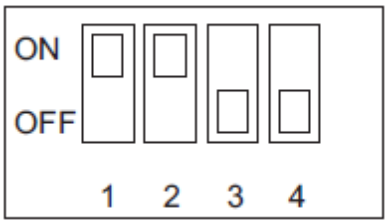


2.5.7 MECHATROLINK-II Settings

(1) Setting Switch SW1 and SW2



SW1 (factory setting)



SW2 (factory settings)

SW2	Function	Setting	Information	Factory Setting
Pin1	Transmission Speed	OFF	4Mbps(MECHATROLINK-I)	ON

	Setting	ON	10 Mbps (MECHATROLINK-II)	
Pin2	Setting the Number of Transmitting Bytes	OFF	17 bytes	ON
		ON	32 bytes	
Pin3	Station Address Setting	OFF	Station address = 40H + SW1	OFF
		ON	Station address = 50H + SW1	
Pin4	Reserved	OFF	-	OFF

(2) Setting the Station Address

Bit3 of SW2	SW1	Station Address
OFF	0	Disabled
OFF	1	41H
OFF	2	42H
OFF	3	43H
OFF	4	44H
OFF	5	45H
OFF	6	46H
OFF	7	47H
OFF	8	48H
OFF	9	49H
OFF	A	4AH
OFF	B	4BH
OFF	C	4CH
OFF	D	4DH
OFF	E	4EH
OFF	F	4FH

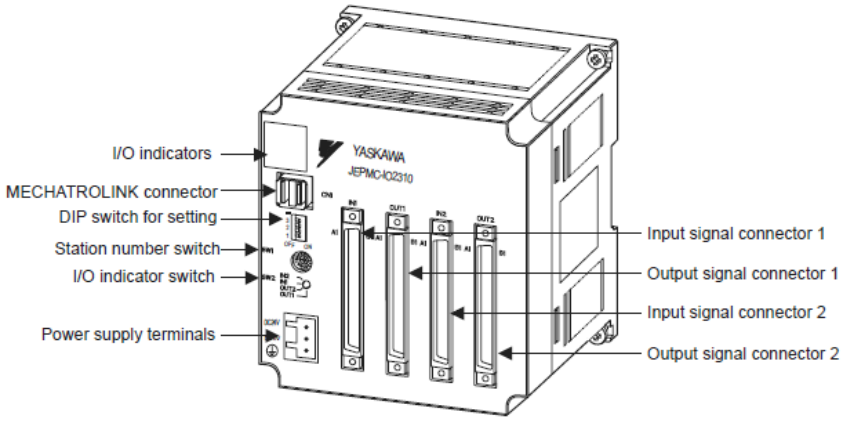
Bit3 of SW2	SW1	Station Address
ON	0	50H
ON	1	51H
ON	2	52H
ON	3	53H
ON	4	54H
ON	5	55H
ON	6	56H
ON	7	57H
ON	8	58H
ON	9	59H
ON	A	5AH
ON	B	5BH
ON	C	5CH
ON	D	5DH
ON	E	5EH
ON	F	5FH

3. Simple I/O Module

3.1 64 I/O Module(IO2310)

3.1.1 Dimension and Configuration

(1) IO2310 Module

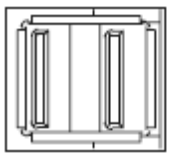


(2) I/O and LED Status

R	ACTIVE		F
1	9	17	25
2	10	18	26
3	11	19	27
4	12	20	28
5	13	21	29
6	14	22	30
7	15	23	31
8	16	24	32

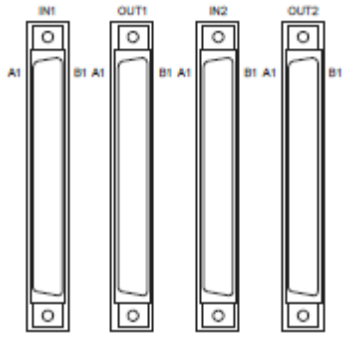
Indication	Color	Meaning
R	Yellow	Not used.
ACTIVE	Yellow	Transmitting data.
F	Red	Fuse is disconnected
1 ~ 32	Yellow	Signal Detection of either input or output

(3) MECHATROLINK Connector



It is used to connect to the MECHATROLINK cable.

(4) I/O Signal Connector



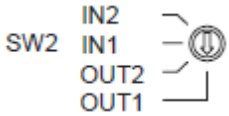
It is used to connect the 64 Input terminals and 64 output terminals.

(5) Station Number Switch



It is used to set the Station Number of MECHATROLINK System. Setting Range is between 0 and E. A single system cannot used duplicate station number.

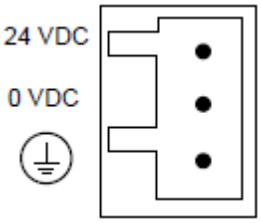
(6) I/O Indication Switch



IN1: Input Signal 1 ~ 32
IN2: Input Signal 33 ~ 64

OUT1: Output Signal 1 ~ 32
OUT2: Output Signal 33 ~ 64

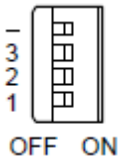
(7) External Connection Terminal



Terminal Name	Function
DC24V	+24VDC
DC0V	0VDC

FG	Grounding

(8) Dip Switch for Setting



Indication	Name	Status	Function	Factory Setting
-	Reserved	-	OFF	OFF
3	MECHATROLINK high-level address setting	ON	7xh	OFF
		OFF	6xh	
2	I/O Byte Setting	ON	32-byte mode	OFF
		OFF	17-byte mode	
1	Communication Speed Setting	ON	10Mbps	ON
		OFF	4Mbps	

(9) Next table shows slave address setting.

Station Address	DIP switch for "3"	Station Number Switch	Station Address	DIP switch for "3"	Station Number Switch
1(61h)	OFF	1	16(70h)	ON	0
2(62h)	OFF	2	17(71h)	ON	1
3(63h)	OFF	3	18(72h)	ON	2
4(64h)	OFF	4	19(73h)	ON	3
5(65h)	OFF	5	20(74h)	ON	4
6(66h)	OFF	6	21(75h)	ON	5
7(67h)	OFF	7	22(76h)	ON	6
8(68h)	OFF	8	23(77h)	ON	7
9(69h)	OFF	9	24(78h)	ON	8
10(6Ah)	OFF	A	25(79h)	ON	9
11(6Bh)	OFF	B	26(7Ah)	ON	A
12(6Ch)	OFF	C	27(7Bh)	ON	B
13(6Dh)	OFF	D	28(7Ch)	ON	C
14(6Eh)	OFF	E	29(7Dh)	ON	D
15(6Fh)	OFF	F	Not used	ON	E, F

3.1.2 Performance Specification

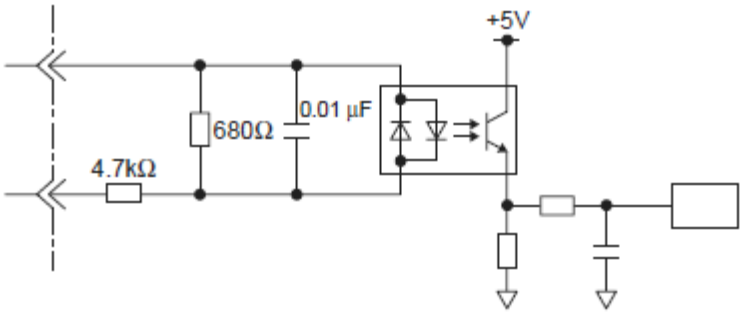
Next table show the performance specification of IO2310.

Item	Specification
Name	64-point I/O Module
Model Name	IO2310
Model Number	JEPMC-IO2310
External Power	24VDC(20.4 ~ 28.8VDC)
Rated Voltage	0.5A
Influx Voltage	1A
Dimension	120 x 130 x 105 (W x H x D)

(1) Input Circuit

Next table shows the specification of input circuit.

Item	specification
Number of Input Terminals	64
Input Form	Sinking or Sourcing
Dielectricity Method	Photo coupler
Input Voltage	24VDC(20.4 ~ 28.8VDC)
Input Current	5mA/point
On Voltage/Current	9V min/1.6mA min
OFF Voltage/Current	7V max/1.3mA max
ON Time / OFF Time	ON time: 2ms, OFF time: 3ms
Number of Input Terminals per Common	16

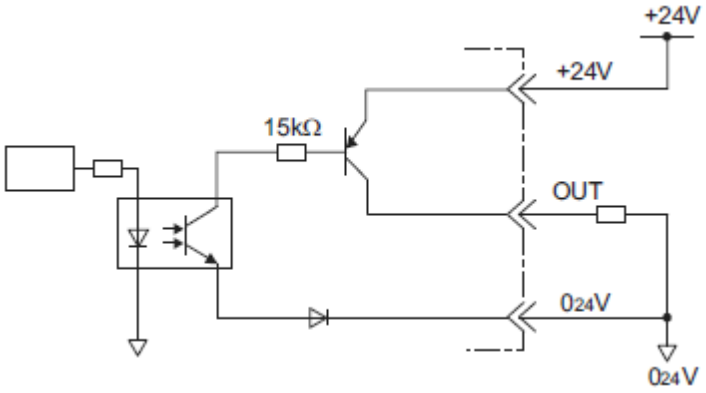


Input Circuit

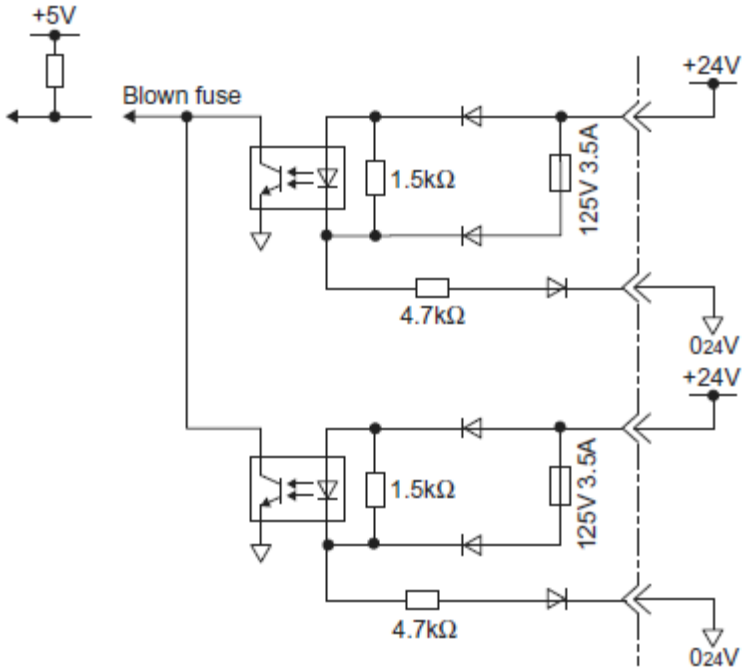
(2) Output Circuit

Next table shows the specification of output circuit.

Item	Specification
Module	IO2310
Number of output terminals	64
Output Form	Transistor, open collector, or sourcing
Dielectricity Method	Photo coupler
Output Voltage	24VDC(20.4 ~ 28.8VDC)
Output Current	50mA/point
Leakage Current in OFF	0.1mA max
ON Time / OFF Time	On time: 2ms max, OFF time: 4ms max
Output Terminals per Common	16
Fuses	Each common terminal is protected by the fuses from the output circuit in case of short-circuiting.
Error Detection	Detect blown fuses



Output Circuit



Blow Fuse Detection Circuit

3.1.3 System Connection

(1) Connector Specification

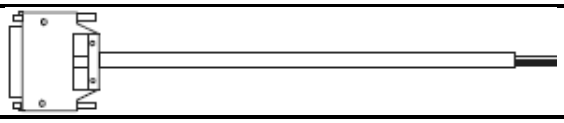
Next Table shows the Connector specification.

Name	Pin No.	Specification			
		Module Side	Producer	Cable Side	Producer
I/O connector	40	900413-1	Tyco Electronics AMP K K	FCN-360C-040-E(cover) FCN-361J-040-AU	Fujitsu Component Limited

(2) Standard Cable and Form

Next table shows the standard cable and form.

Name	Model	Length (m)	Form (JEPMC-W5410-□□)
I/O Cable	JEPMC-W5410-05	0.5	
	JEPMC-W5410-10	1	
	JEPMC-W5410-30	3	

(3) Chart on Lines of Standard Cable

Next Table shows the Lines of Standard Cable (JEPMC-W5410-□□).

No.	Color	Terminal Indication Color	Terminal Indication
A1	Blue	Red	-
B1	Blue	Black	-
A2	Pink	Red	-
B2	Pink	Black	-
A3	Green	Red	-
B3	Green	Black	-
A4	Orange	Red	-
B4	Orange	Black	-
A5	Gray	Red	-
B5	Gray	Black	-
A6	Blue	Red	-
B6	Blue	Black	-
A7	Pink	Red	-
B7	Pink	Black	-
A8	Green	Red	-
B8	Green	Black	-
A9	Orange	Red	-
B9	Orange	Black	-
A10	Gray	Red	-
B10	Gray	Black	-

No.	Color	Terminal Indication Color	Terminal Indication
A11	Blue	Red	---
B11	Blue	Black	---
A12	Pink	Red	---
B12	Pink	Black	---
A13	Green	Red	---
B13	Green	Black	---
A14	Orange	Red	---
B14	Orange	Black	---
A15	Gray	Red	---
B15	Gray	Black	---
A16	Blue	Red	---
B16	Blue	Black	---
A17	Pink	Red	---
B17	Pink	Black	---
A18	Green	Red	---
B18	Green	Black	---
A19	Orange	Red	---
B19	Orange	Black	---
A20	Gray	Red	---
B20	Gray	Black	---

(4) Connector Pin Configuration

(a) Input Signal ConnectorIN1

No.	Signal Name	Caution	No.	Signal Name	Caution
A1	(NC)		B1	(NC)	
A2	+24V_2	24-V power supply 2	B2	+24V_2	24-V power supply 2
A3	IN32	Input 32	B3	IN31	Input31

A4	IN30	Input 30	B4	IN31	Input29
A5	IN28	Input 28	B5	IN31	Input27
A6	IN26	Input 26	B6	IN31	Input25
A7	IN24	Input 24	B7	IN31	Input23
A8	IN22	Input 22	B8	IN31	Input21
A9	IN20	Input 20	B9	IN31	Input19
A10	IN18	Input 18	B10	IN31	Input17
A11	IN16	Input 16	B11	IN31	Input15
A12	IN14	Input 14	B12	IN31	Input13
A13	IN12	Input 12	B13	IN31	Input11
A14	IN10	Input 10	B14	IN31	Input9
A15	IN08	Input 8	B15	IN31	Input7
A16	IN06	Input 6	B16	IN31	Input5
A17	IN04	Input 4	B17	IN31	Input3
A18	IN02	Input 2	B18	IN31	Input1
A19	(NC)		B19	(NC)	
A20	+24V_1	24-V power supply 1	B20	+24V_1	24-V power supply 1

(b) Input Signal ConnectorIN2

No.	Signal Name	Caution	No.	Signal Name	Caution
A1	(NC)		B1	(NC)	
A2	+24V_4	24-V power supply 4	B2	+24V_4	24-V power supply 4
A3	IN64	Input 32	B3	IN63	Input31
A4	IN62	Input 30	B4	IN61	Input29
A5	IN60	Input 28	B5	IN59	Input27
A6	IN58	Input 26	B6	IN57	Input25
A7	IN56	Input 24	B7	IN55	Input23
A8	IN54	Input 22	B8	IN53	Input21
A9	IN52	Input 20	B9	IN51	Input19
A10	IN50	Input 18	B10	IN49	Input17
A11	IN48	Input 16	B11	IN47	Input15
A12	IN46	Input 14	B12	IN45	Input13
A13	IN44	Input 12	B13	IN43	Input11
A14	IN42	Input 10	B14	IN41	Input9
A15	IN40	Input 8	B15	IN39	Input7
A16	IN38	Input 6	B16	IN37	Input5

A17	IN36	Input 4	B17	IN35	Input3
A18	IN34	Input 2	B18	IN33	Input1
A19	(NC)		B19	(NC)	
A20	+24V_3	24-V power supply 3	B20	+24V_3	24-V power supply 3

(c) Output Signal ConnectorOUT1

No.	Signal Name	Caution	No.	Signal Name	Caution
A1	024V_6	Common ground 6	B1	024V_6	Common ground 6
A2	+24V_2	24-V power supply 6	B2	+24V_2	24-V power supply 6
A3	OUT32	Output 32	B3	OUT31	Output31
A4	OUT 30	Output 30	B4	OUT31	Output29
A5	OUT 28	Output 28	B5	OUT31	Output27
A6	OUT26	Output 26	B6	OUT31	Output25
A7	OUT24	Output 24	B7	OUT31	Output23
A8	OUT22	Output 22	B8	OUT31	Output21
A9	OUT20	Output 20	B9	OUT31	Output19
A10	OUT18	Output 18	B10	OUT31	Output17
A11	OUT16	Output 16	B11	OUT31	Output15
A12	OUT14	Output 14	B12	OUT31	Output13
A13	OUT12	Output 12	B13	OUT31	Output11
A14	OUT10	Output 10	B14	OUT31	Output9
A15	OUT08	Output 8	B15	OUT31	Output7
A16	OUT06	Output 6	B16	OUT31	Output5
A17	OUT04	Output 4	B17	OUT31	Output3
A18	OUT02	Output 2	B18	OUT31	Output1
A19	024V_5	Common ground 5	B19	024V_5	Common ground 5
A20	+24V_1	24-V power supply 5	B20	+24V_1	24-V power supply 5

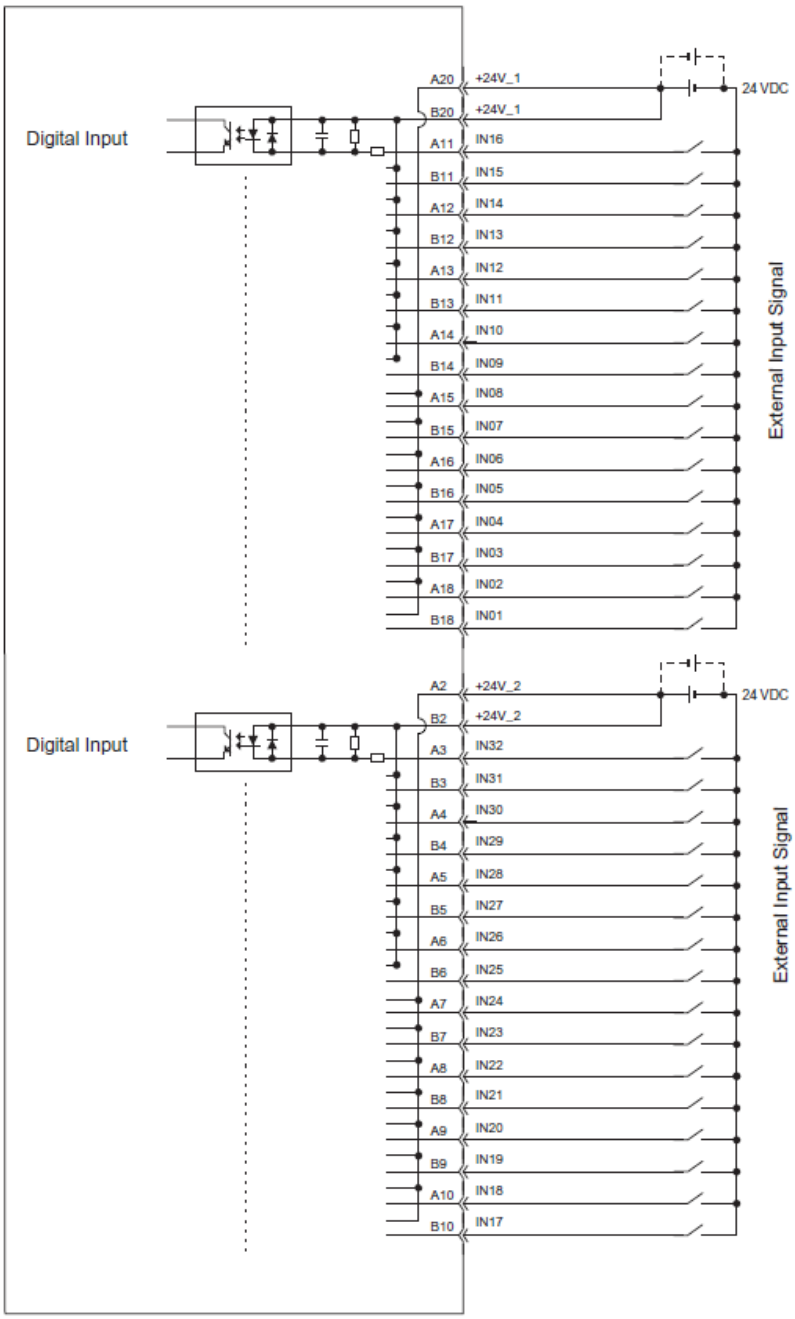
(d) Output Signal ConnectorOUT2

No.	Signal Name	Caution	No.	Signal Name	Caution
A1	024V_8	Common ground 8	B1	024V_8	Common ground 8
A2	+24V_4	24-V power supply 4	B2	+24V_4	24-V power supply 4
A3	OUT64	Output 32	B3	OUT63	Output31
A4	OUT62	Output 30	B4	OUT61	Output29
A5	OUT60	Output 28	B5	OUT59	Output27

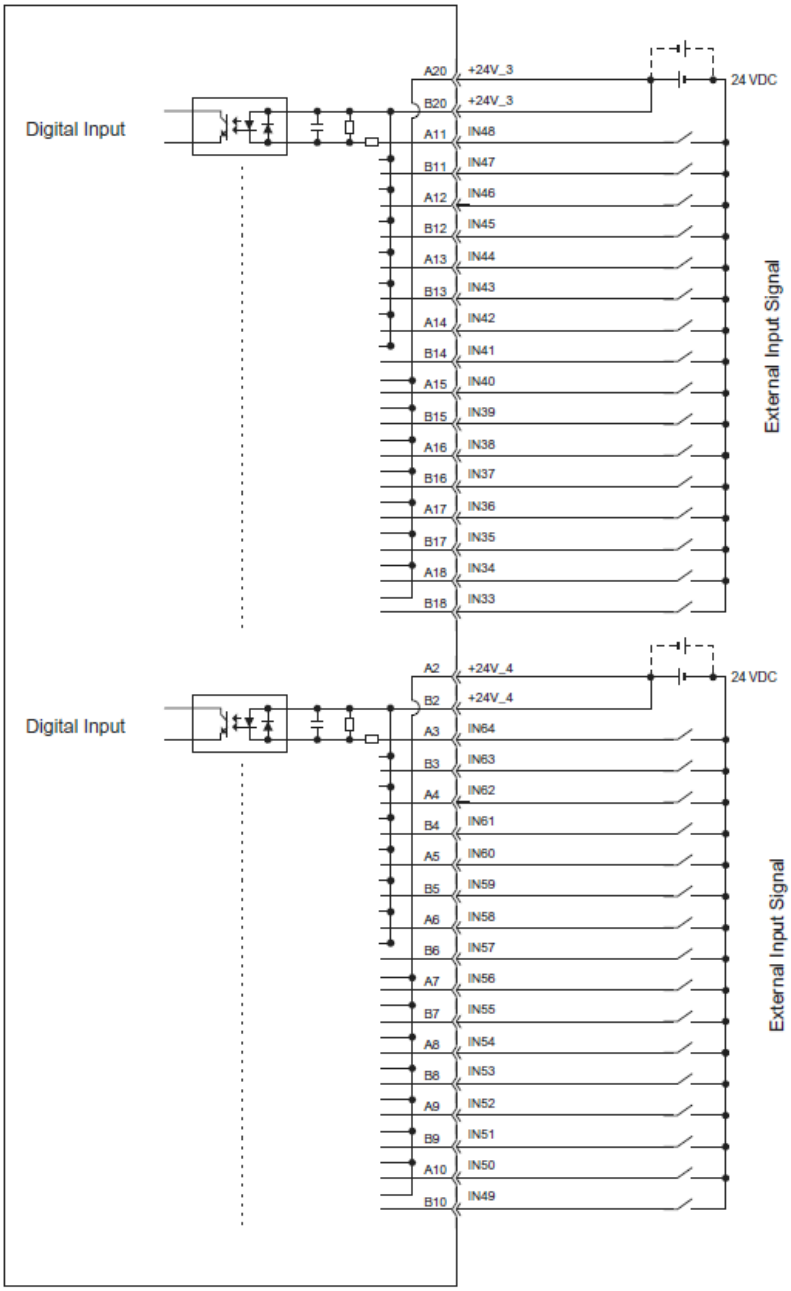
A6	OUT58	Output 26	B6	OUT57	Output25
A7	OUT56	Output 24	B7	OUT55	Output23
A8	OUT54	Output 22	B8	OUT53	Output21
A9	OUT52	Output 20	B9	OUT51	Output19
A10	OUT50	Output 18	B10	OUT49	Output17
A11	OUT48	Output 16	B11	OUT47	Output15
A12	OUT46	Output 14	B12	OUT45	Output13
A13	OUT44	Output 12	B13	OUT43	Output11
A14	OUT42	Output 10	B14	OUT41	Output9
A15	OUT40	Output 8	B15	OUT39	Output7
A16	OUT38	Output 6	B16	OUT37	Output5
A17	OUT36	Output 4	B17	OUT35	Output3
A18	OUT34	Output 2	B18	OUT33	Output1
A19	024V_7	Common ground 7	B19	024V_7	Common ground 7
A20	+24V_3	24-V power supply 3	B20	+24V_3	24-V power supply 3

(5) Connection Example

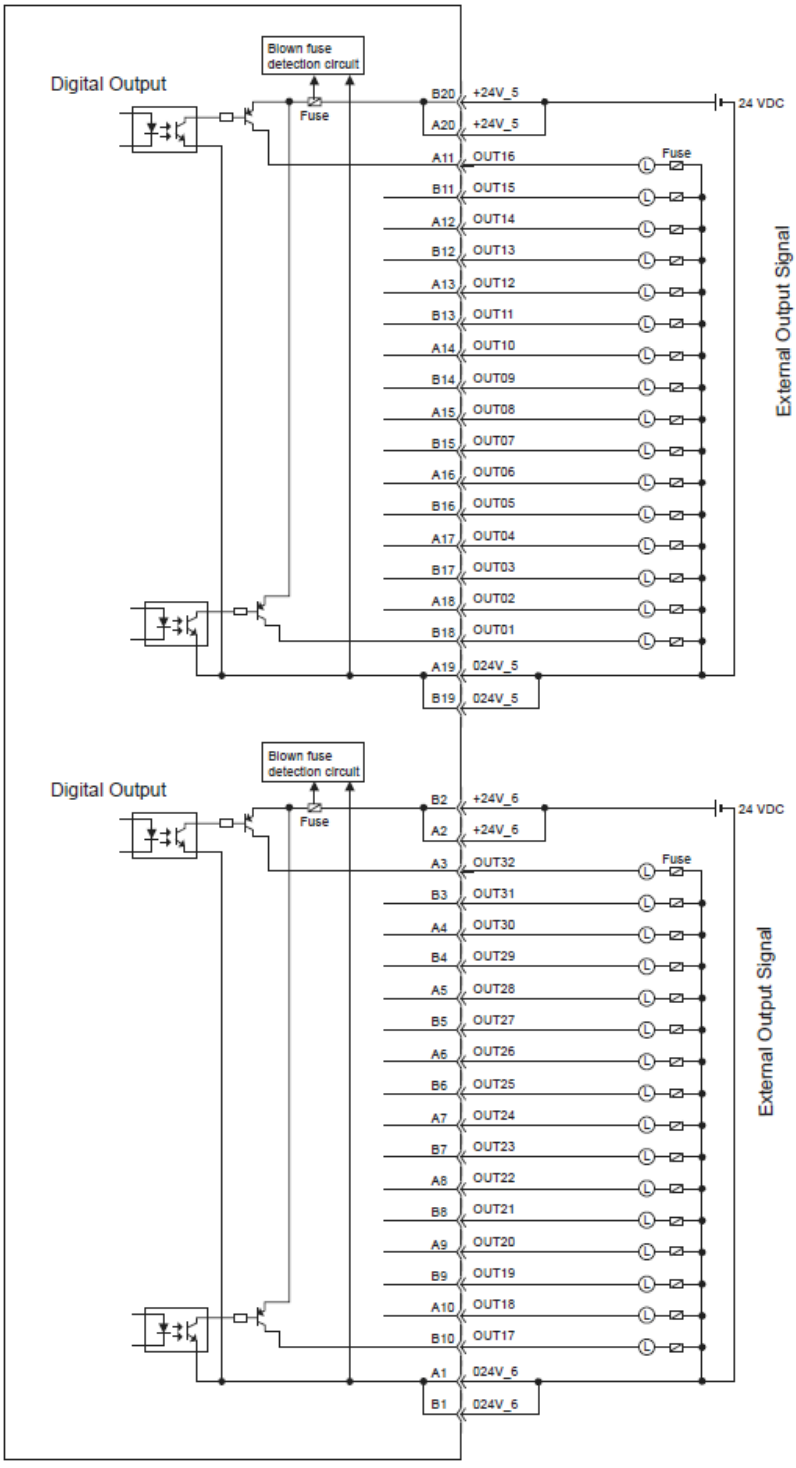
(a) Input Signal ConnectorIN1



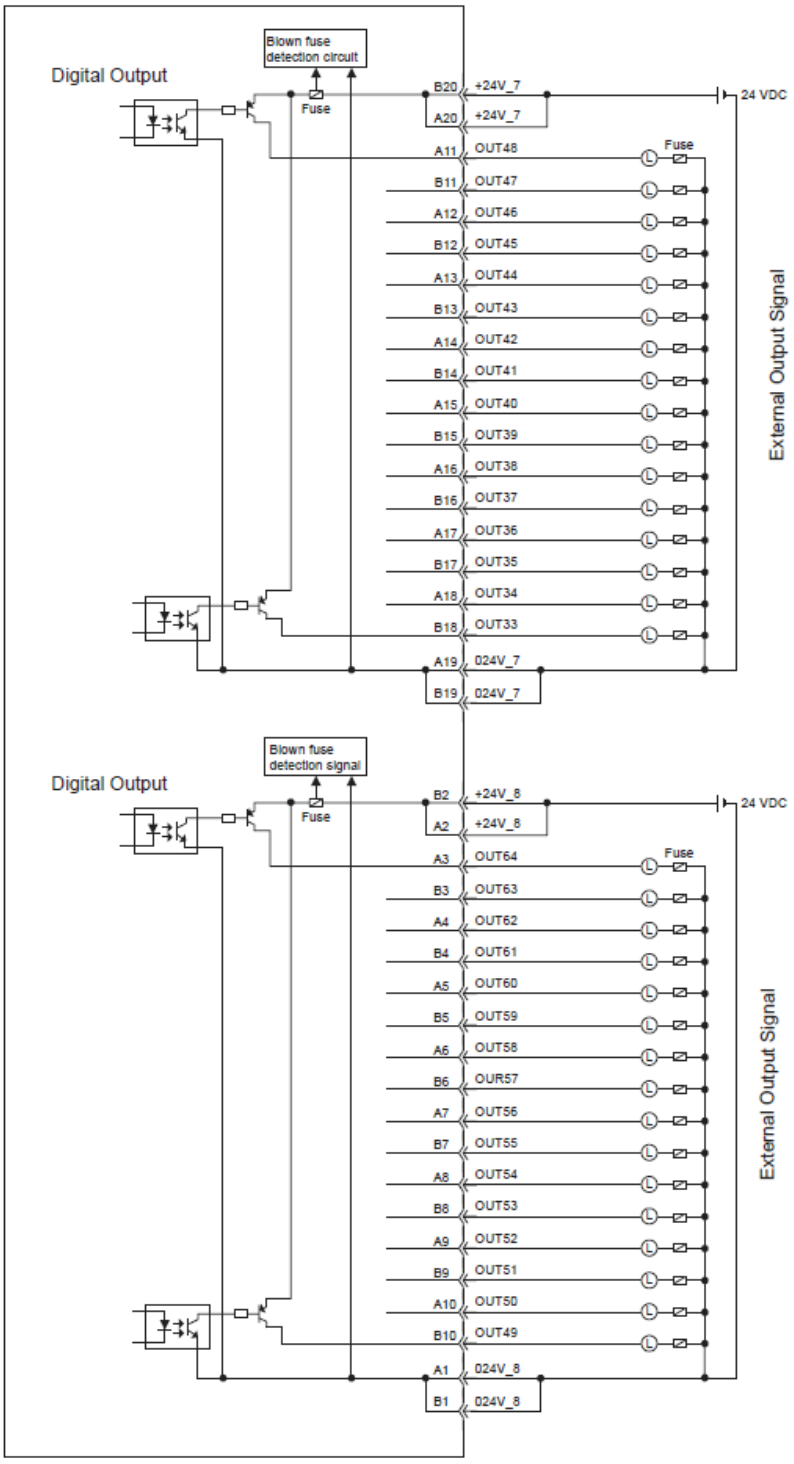
(b) Input Signal ConnectorIN2



(c) Output Signal OUT1



(d) Output Signal ConnectorOUT2

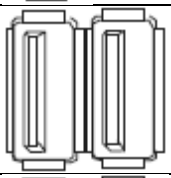
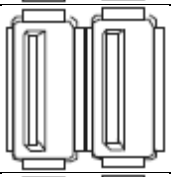
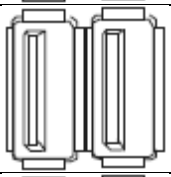
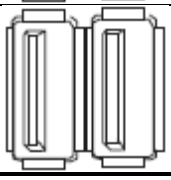


4. Connections

4.1 MECHATROLINK Connectors

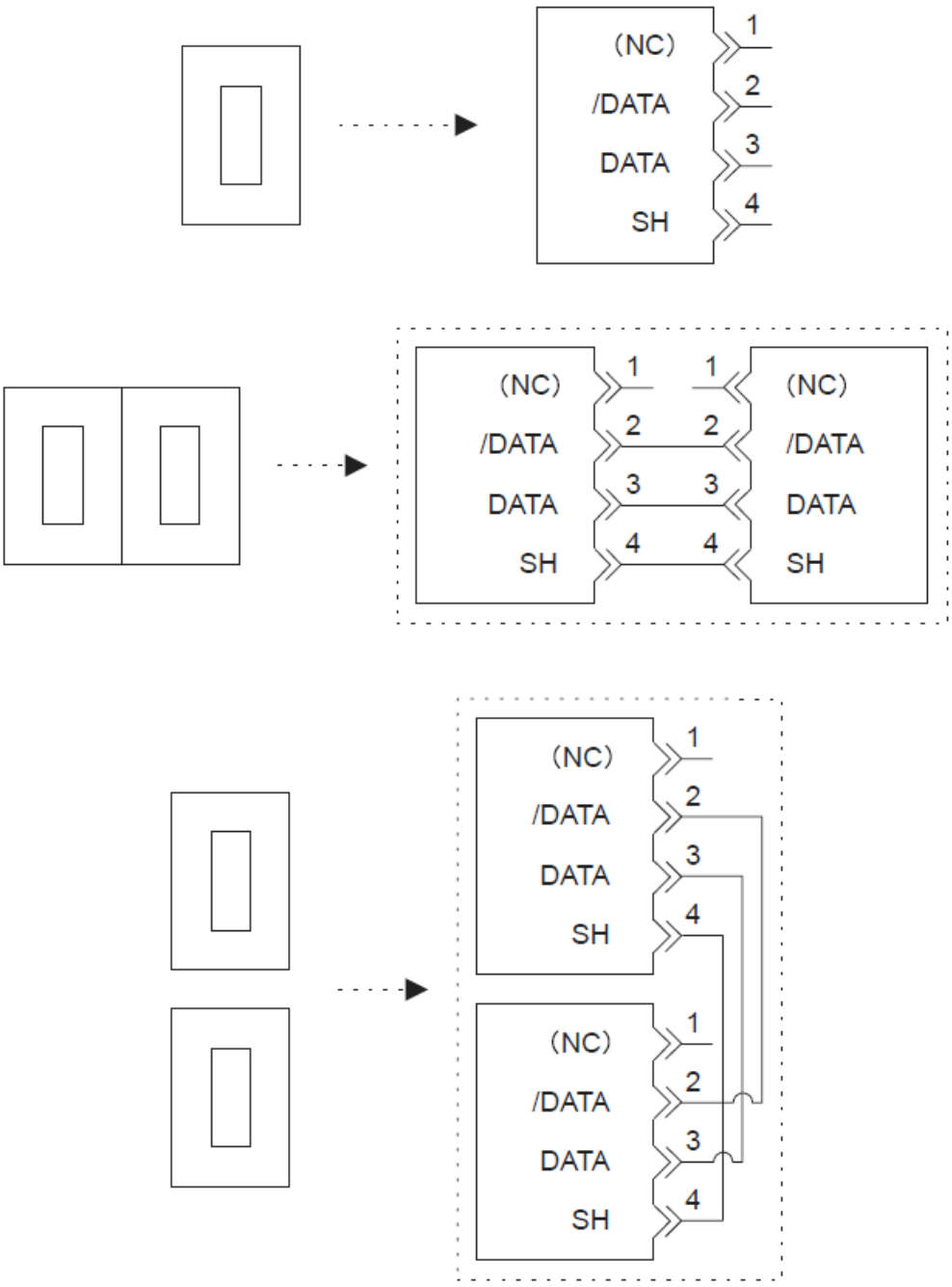
(1) Connector Type

Modules of MECHATROLINK Connector are shown in the next table.

Module	Number of Connectors	Form	Connector Name
PCI R1604MLII	2		NET#1
			NET#2
JEPMC-AN2900	1		CN1
JEPMC-AN2910	1		CN1
JEPMC-PL2900	1		CN1
JEPMC-IO2310	1		CN1

(2) Internal Connections

Next diagram shows the internal connections of MECHATROLINK Connector.

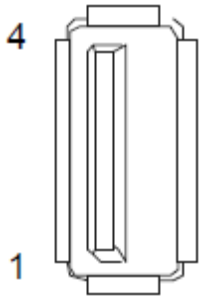


USD Terminator (JEPMC-W6022) must be installed at the end of MECHATROLINK system.

(3) Connector Specification

Next shows Connector Specification.

Name	Pin No.	Connector Model		
		Module Connector	Cable Connector	Manufacturer
MECHATROLINK-Connectors	4	USB-ARA41B1-C50	DUSB-APA41B1-C50	DDK



No.	Signal Name	Information
1	(NC)	NOT USED
2	/DATA	Signal minus(-)
3	DATA	Signal plus(+)
4	SH	NOT USED
Shell	Shield	Connect with Shield Cable

4.2 MECHATROLINK Cables

Next shows the Standard Cable List produced by Yaskawa.

(1) Standard Cable List

Cable Name and Specifications	Model	Length (m)
MECHATROLINK Cable USB connector to USB connector	JEPMC-W6002-A5	0.5
	JEPMC-W6002-01	1
	JEPMC-W6002-03	3
	JEPMC-W6002-05	5
	JEPMC-W6002-10	10
	JEPMC-W6002-20	20
	JEPMC-W6002-30	30
	JEPMC-W6002-40	40
	JEPMC-W6002-50	50
MECHATROLINK Cable USB connector to USB connector (with ferrite core)	JEPMC-W6003-A5	0.5
	JEPMC-W6003-01	1
	JEPMC-W6003-03	3
	JEPMC-W6003-05	5
	JEPMC-W6003-10	10
	JEPMC-W6003-20	20
	JEPMC-W6003-30	30
	JEPMC-W6003-40	40
	JEPMC-W6003-50	50
MECHATROLINK Cable USB connector to loose wires	JEPMC-W6011-A5	0.5
	JEPMC-W6011-01	1

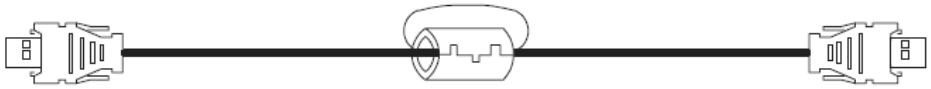
	JEPMC-W6011-03	3
	JEPMC-W6011-05	5
	JEPMC-W6011-10	10
	JEPMC-W6011-20	20
	JEPMC-W6011-30	30
	JEPMC-W6011-40	40
	JEPMC-W6011-50	50
Terminator (Terminating resistor) 130 Ω	JEPMC-W6022	-

(2) Cable Form

(a) Model: JEPMC-W6002-□□



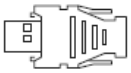
(b) Model: JEPMC-W6003-□□

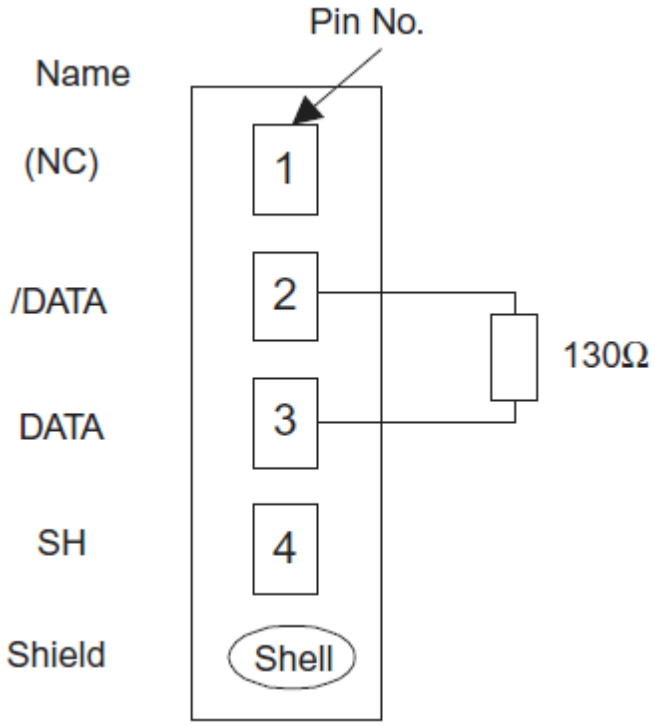


(c) Model: JEPMC-W6011-□□



(d) Model: JEPMC-W6022

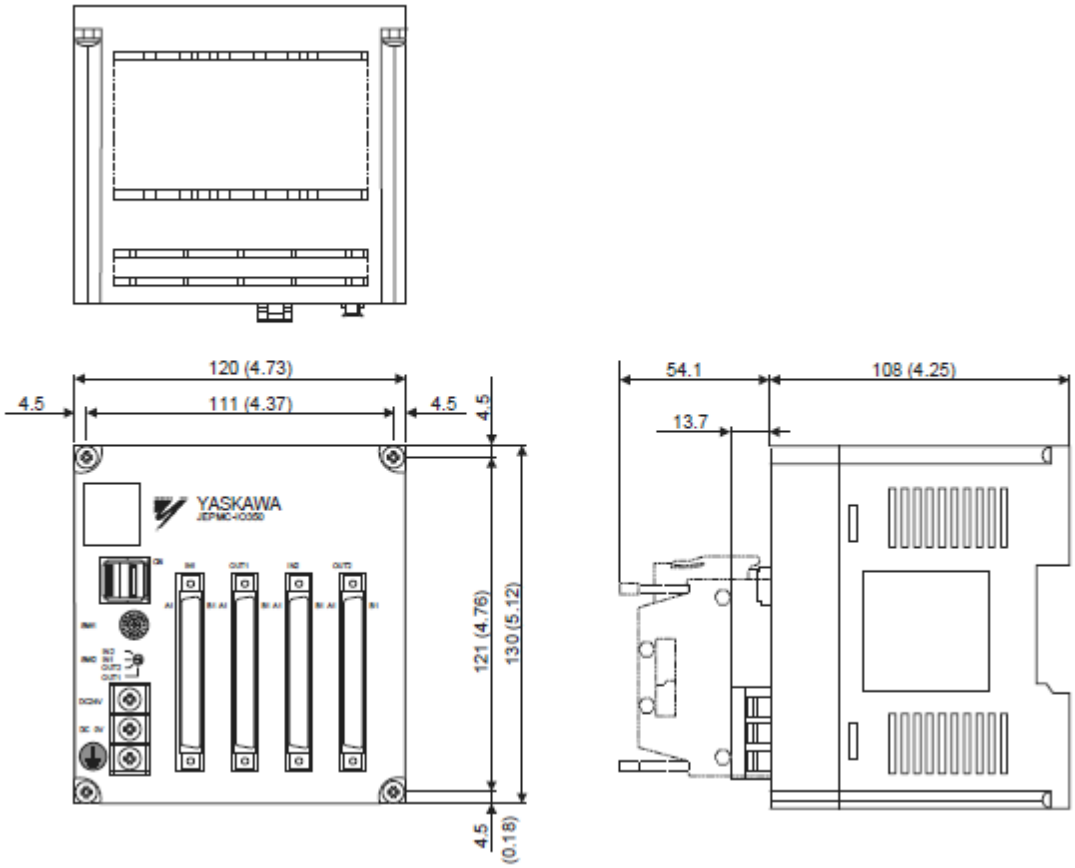




5. Appendix A Dimension

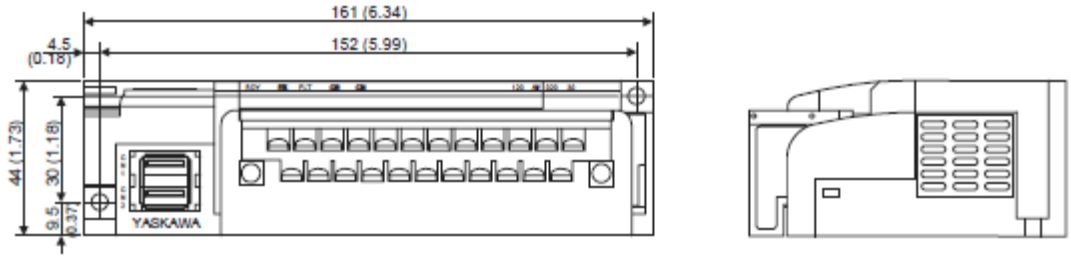
A.1 I/O Modules

A.1.1 64 Point I/O Module

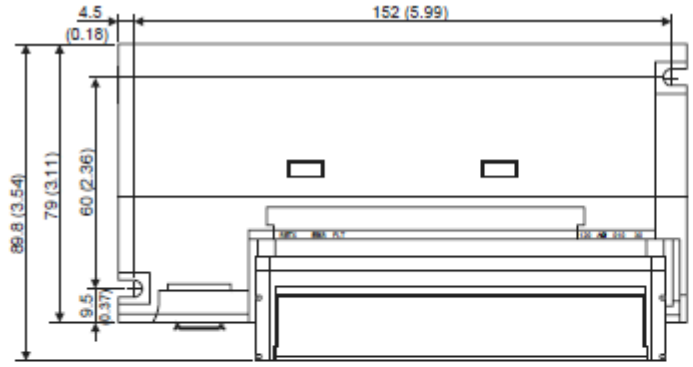


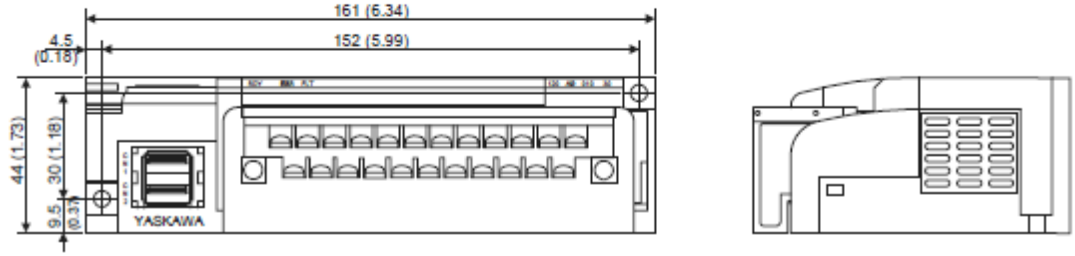
A.1.2 Analog Input Module (±10V, 4 Channel)



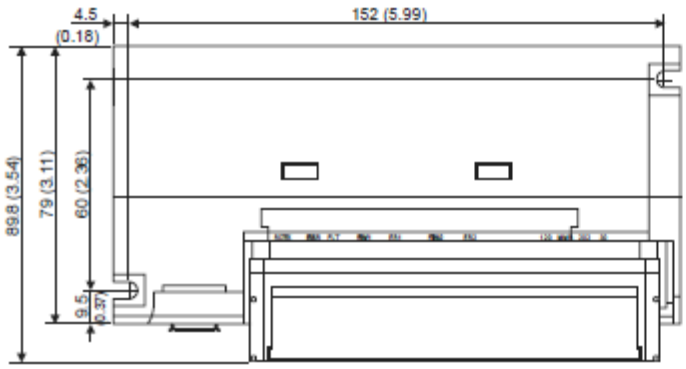


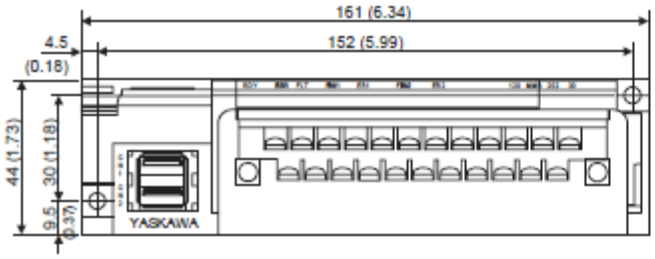
A.1.3 Analog Input Module (±10V, 2 Channel)

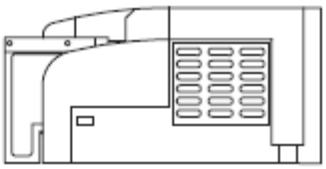




A.2 Pulse Output Module


4.5
(0.18)
152 (5.99)
89.8 (3.54)
79 (3.11)
60 (2.36)
9.5
(0.37)


161 (6.34)
4.5
(0.18)
152 (5.99)
44 (1.73)
30 (1.18)
9.5
(0.37)
YASKAWA



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